

Ex.No:1

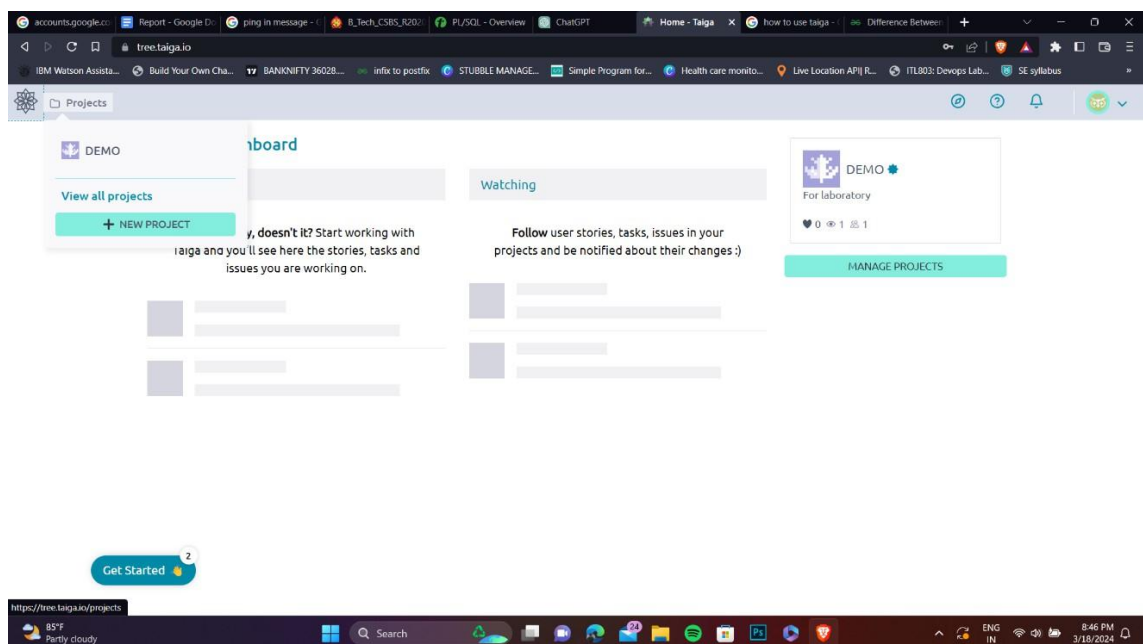
Estimate the IT Project Cost and Control using open-source tools**AIM:****ALGORITHM:****Step 1:**

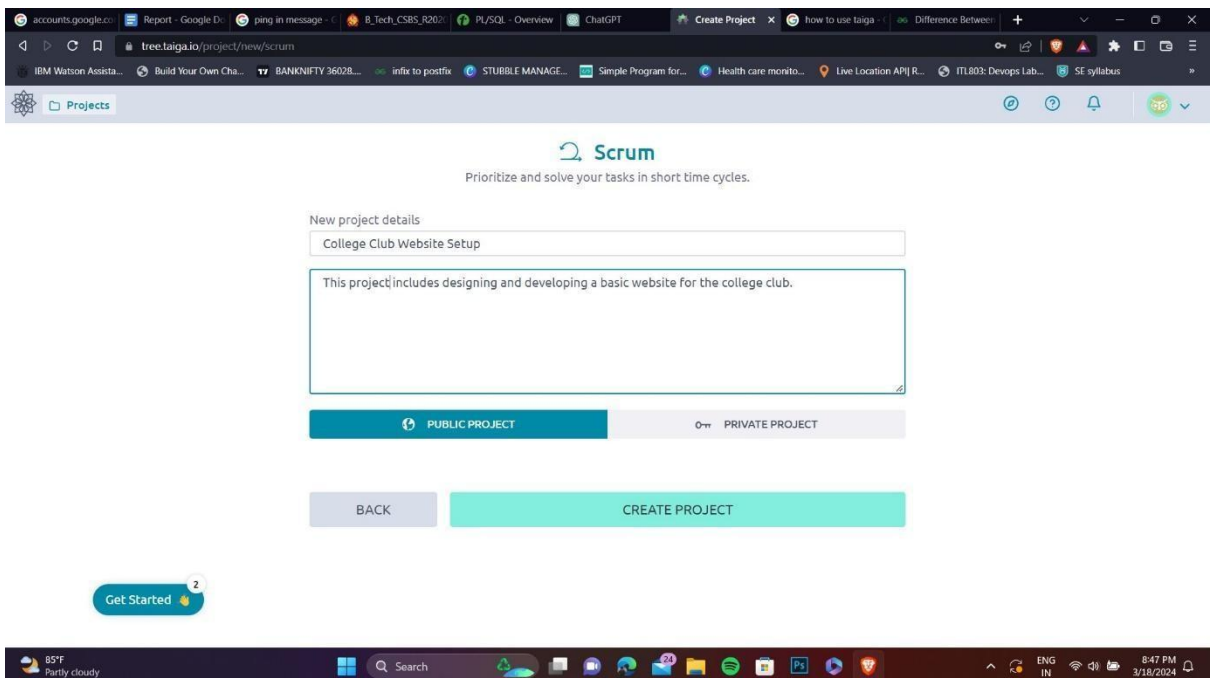
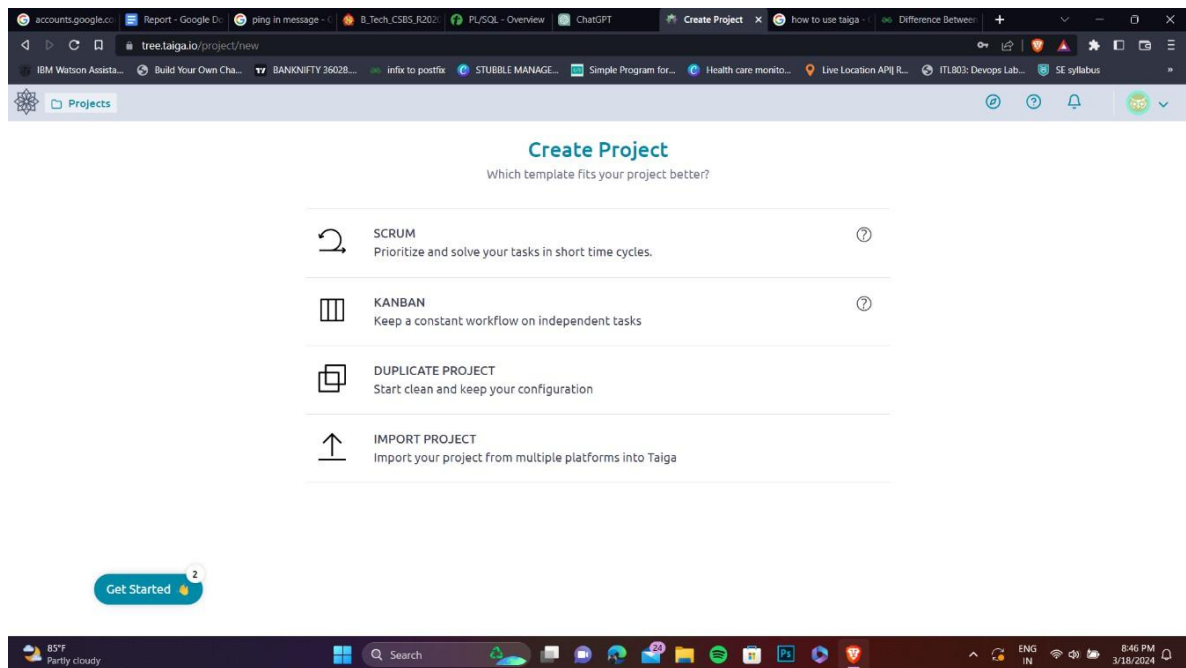
Create an Account in TAIGA open source online tool.

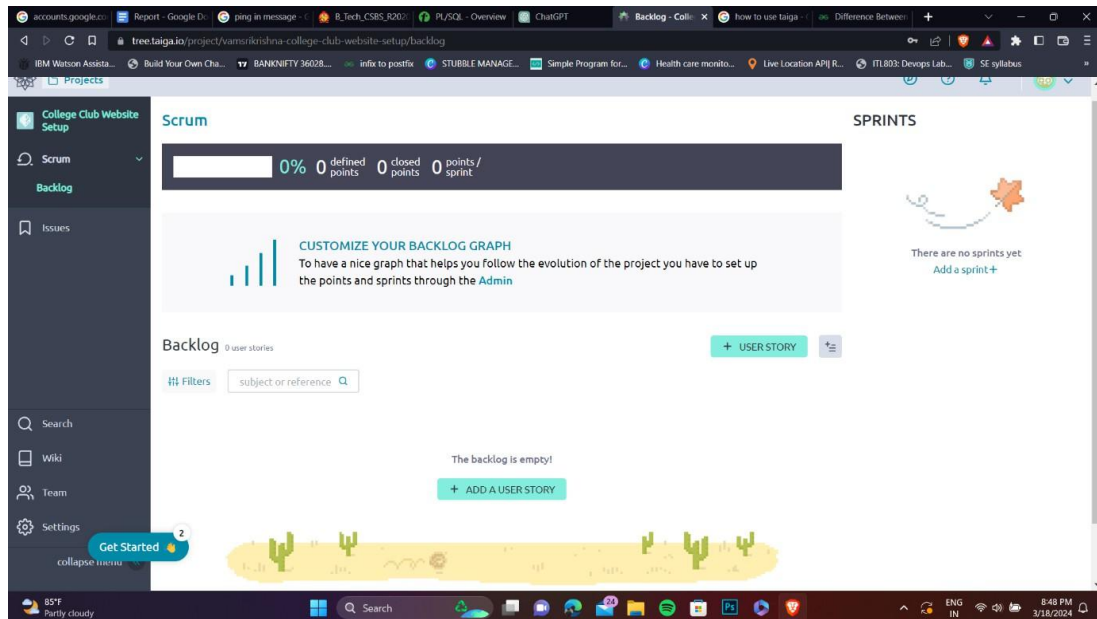
Step 2:

Create a new project in Taiga named "College Club Website Setup."

Define the project's scope, which includes designing and developing a basic website for the college club..







Click Add a User Story

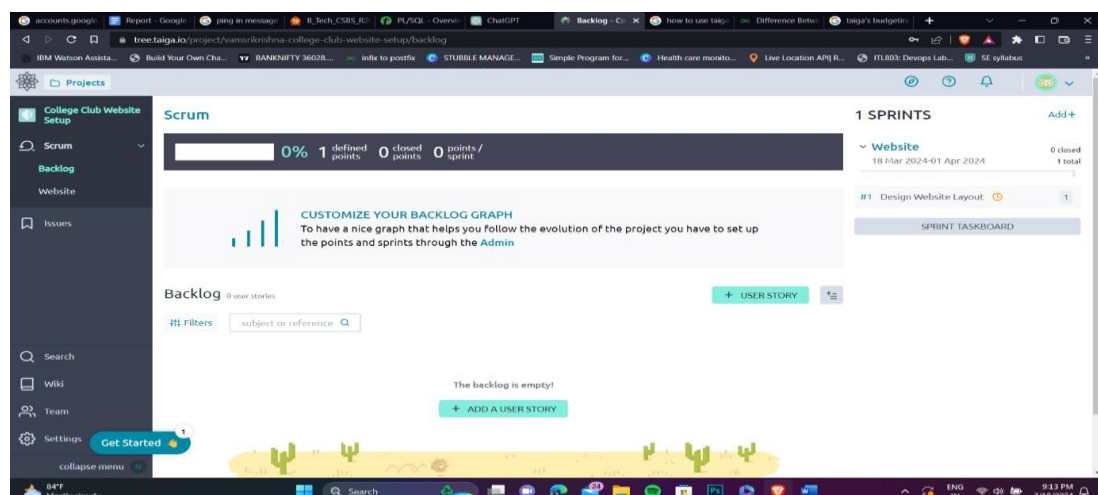
Step 3:

Break down the project into smaller tasks within Taiga's backlog. Tasks may include:

- Gathering requirements
- Designing website layout

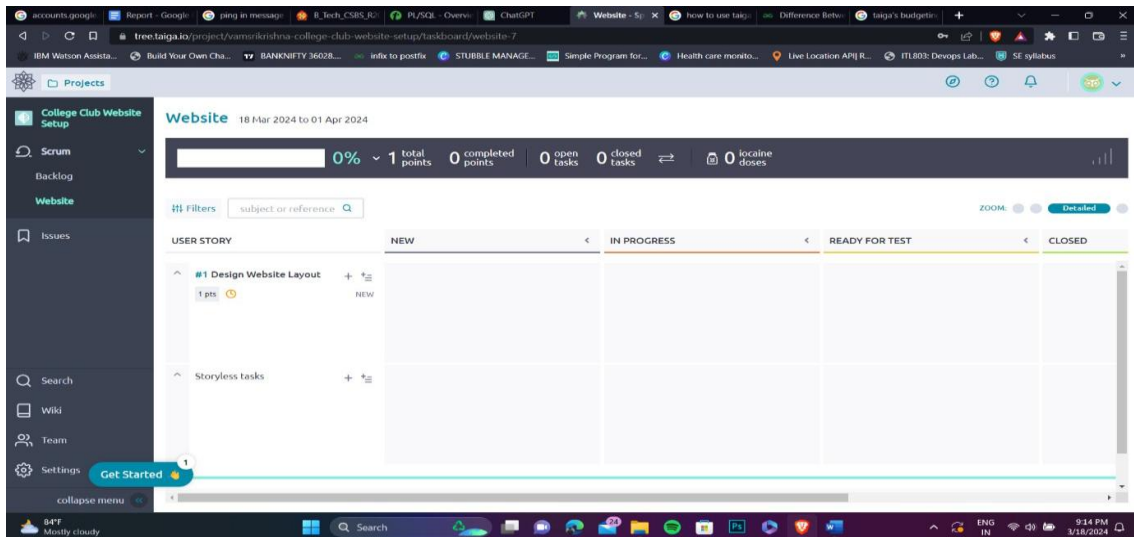
Step 4:

Click Create



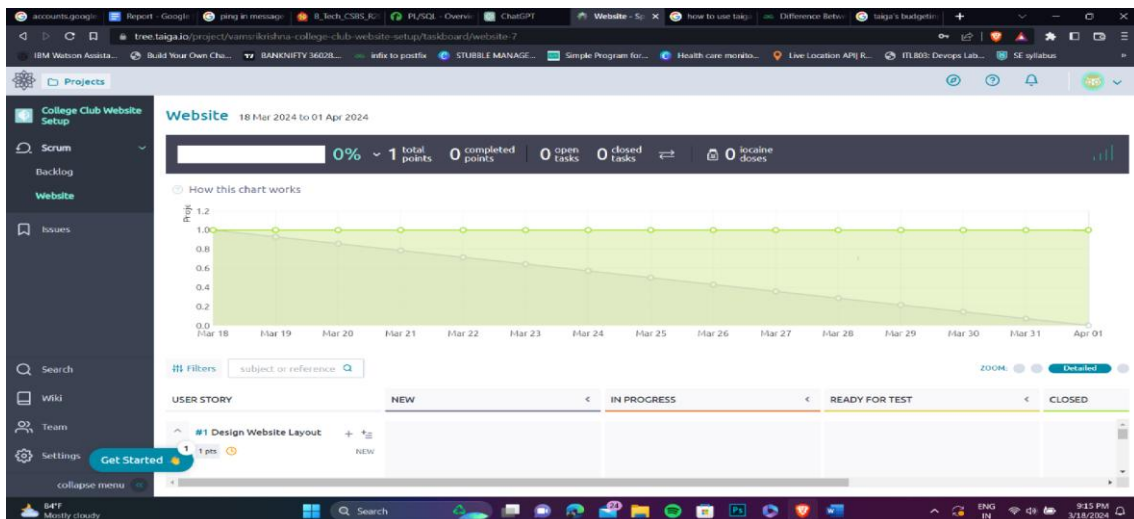
Step 5:

Now drag User Story to Sprint in the right side by creating new sprint as WEBSITE



Now we could able to view the tasks to be done and it's progress.

Now click on the Network icon in the top right, we could able to calculate the remaining time and no. of tasks to be completed in that sprint duration.

**RESULT:**

Ex.No:2

Scheduling a Project with PERT and CPM**AIM:****ALGORITHM:**

1. Identify Activities:- Activities include:- Requirements gathering- Design phase (UI/UX design)- Frontend development- Backend development- Integration testing- User acceptance testing (UAT)- Deployment- Maintenance and updates
2. Sequence Activities:- Activities must be sequenced logically. For example, design cannot begin until requirements are gathered, and testing cannot start until development is completed.
3. Estimate Activity Durations:- Estimate durations for each activity based on historical data, expert judgment, and team input. For instance, design may take 2 weeks, while development may take 6 weeks.
4. Create a Network Diagram:- Develop a network diagram to visualize the sequence of activities and their dependencies. This diagram helps identify the critical path and understand project flow.
5. Calculate Early Start (ES) and Early Finish (EF) Times:- Calculate ES and EF times for each activity based on dependencies and durations. For instance, if design takes 2 weeks, its ES is week 1 and EF is week 2.
6. Determine Late Start (LS) and Late Finish (LF) Times: - Determine LS and LF times for each activity. For instance, if the project deadline is week 10 and design takes 2 weeks, its LS is week 9 and LF is week 10.
7. Calculate Slack or Float:- Calculate slack or float for each activity to identify flexibility in the schedule. Activities on the critical path have zero slack.
8. Identify the Critical Path:- Identify the critical path, which includes activities with zero slack. Any delay in critical path activities will delay the project's overall completion.

9. Perform PERT Analysis:- Use PERT analysis to calculate the expected duration of the project and estimate the probability of completing it within a specified timeframe. Consider optimistic, pessimistic, and most likely estimates for each activity.

10. Monitor and Control the Schedule:- Continuously monitor activity progress and compare it against the planned schedule. Take corrective actions if deviations or delays occur to ensure the project stays on track.

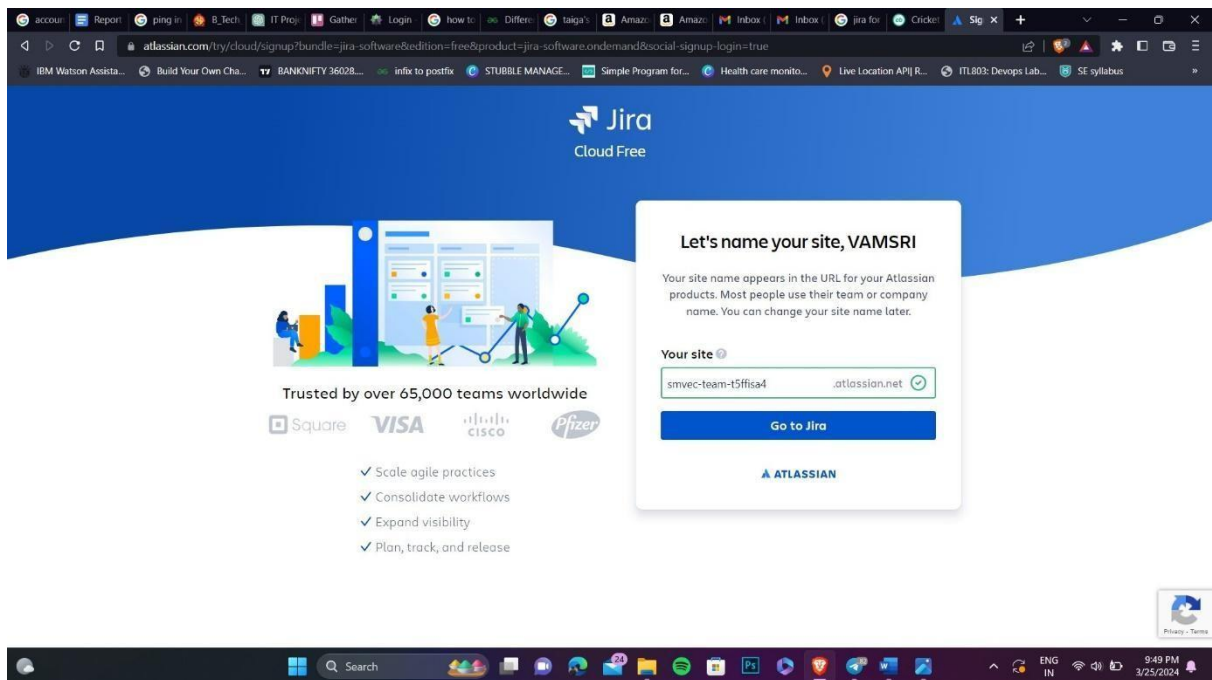
RESULT:

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Ex.No: 3

Estimation of the total time required to complete the project if no delay**AIM:****ALGORITHM:****Step 1:**

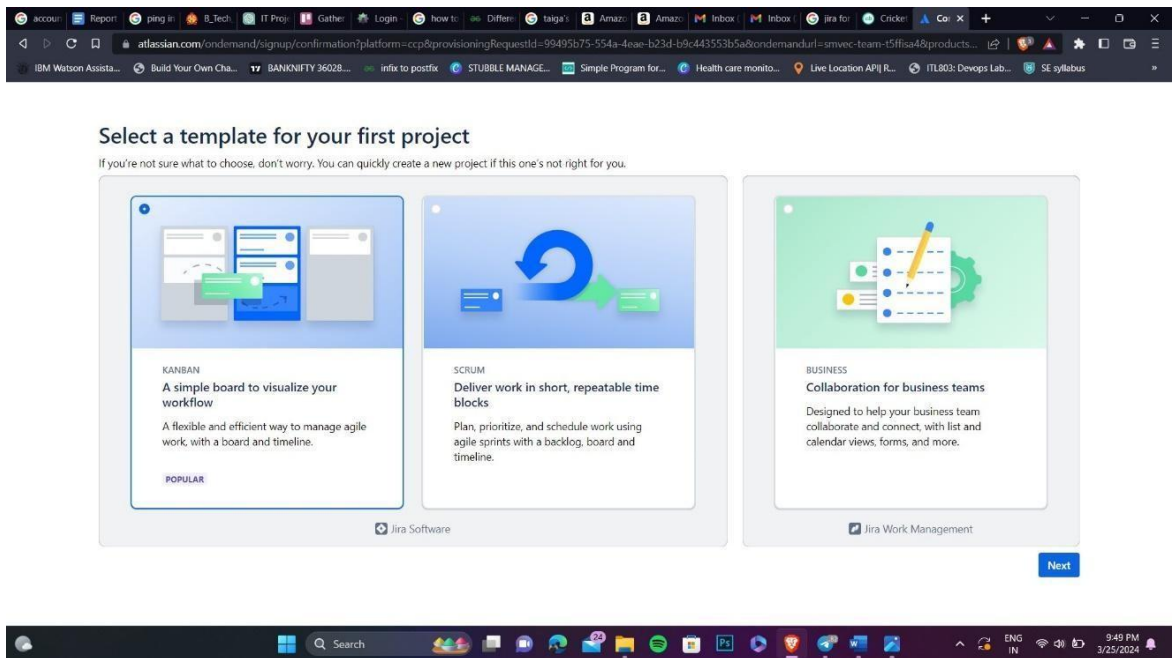
Create a an Account on JIRA Online Tool

<https://www.atlassian.com/try/cloud/signup?bundle=jira-software&edition=free&skipBundles=true>

Now Click Go to JIRA

Step 2:

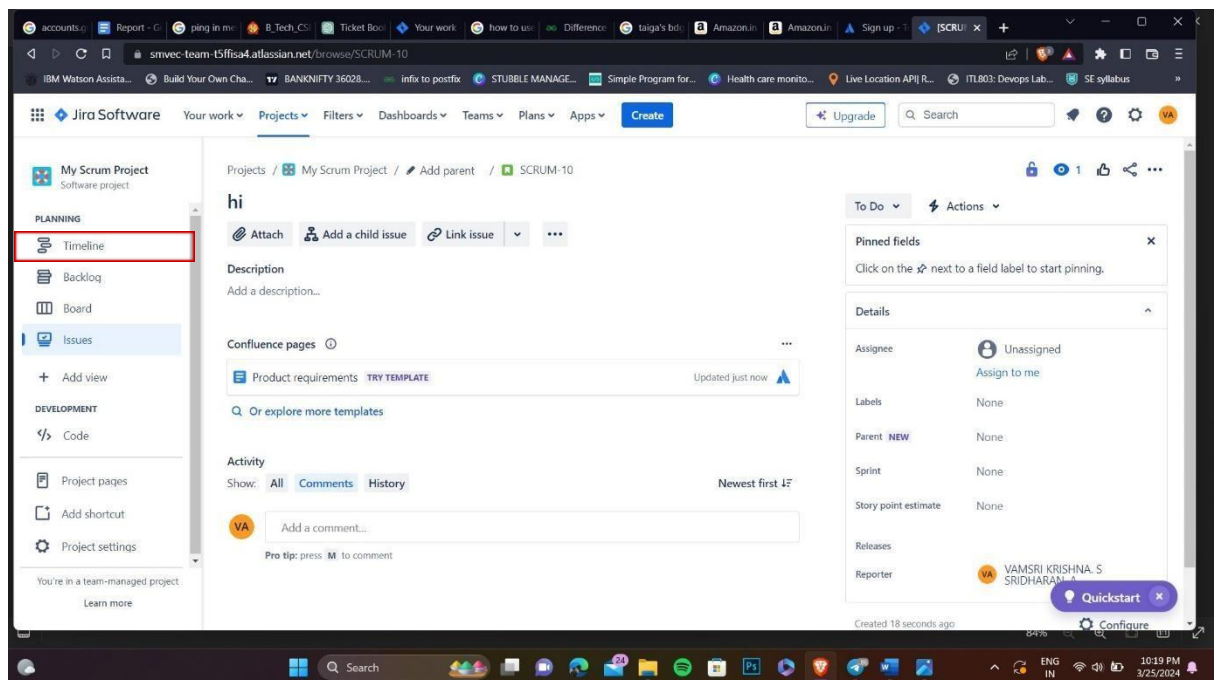
Select SCRUM Below



And Click Next

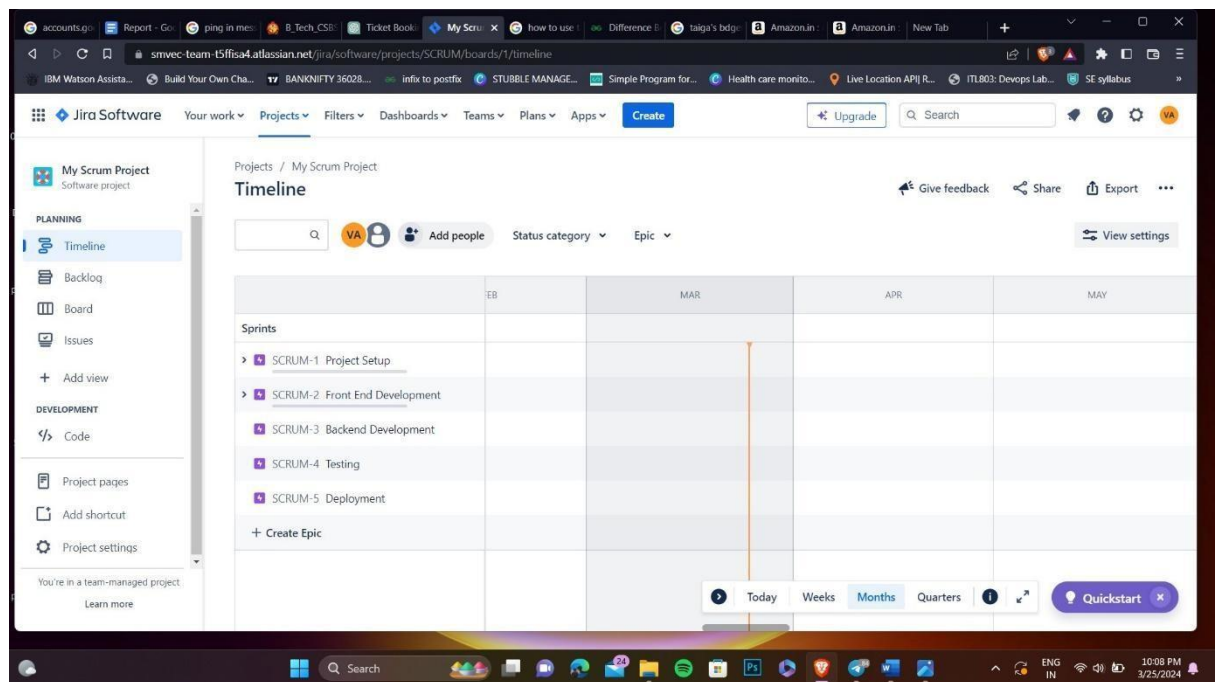
Step 3:

Now We Could View The

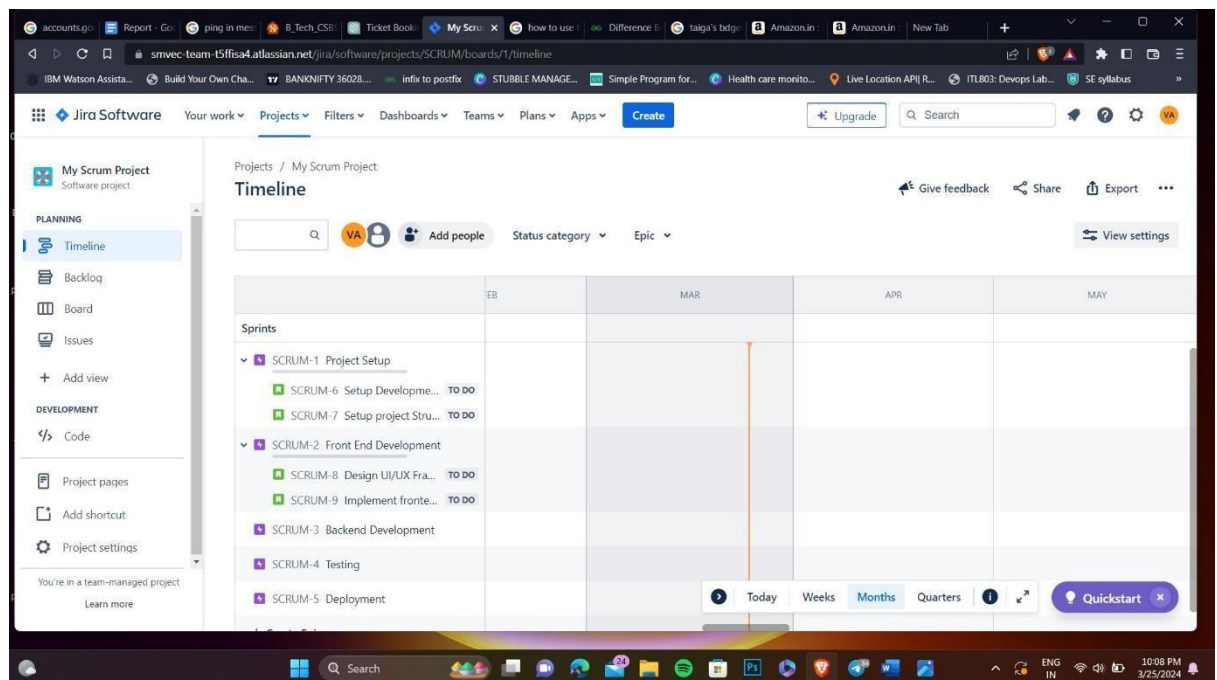


Now Click Timeline in Left

Step 4: By Clicking on the Create Epic Enter the Scrum planning for your project

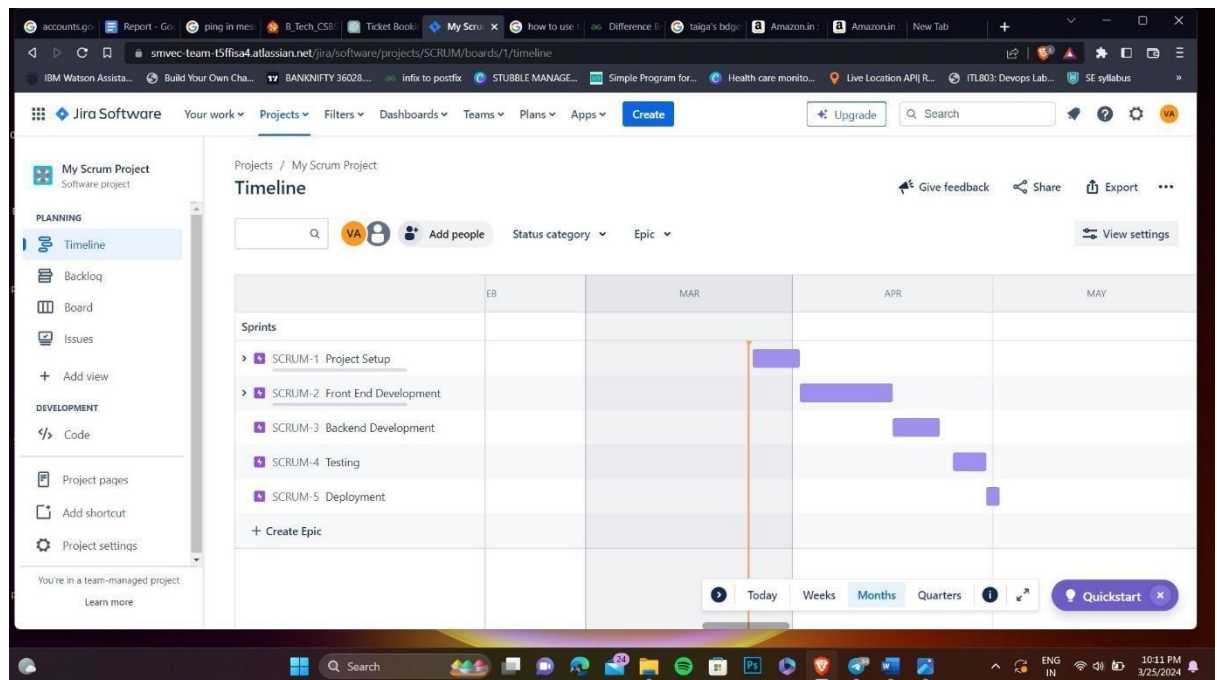


Step 6: Now under Each Scrum Add Create Child Issue by Click on add icon in the scrums



Step 7:

Now Estimate the time Required For The entire Project and assign due for each Scrum by adding the timeline for each SCRUMS



Here we could clearly see the timeline starts from March end to April end.

RESULT:

Ex.No: 4

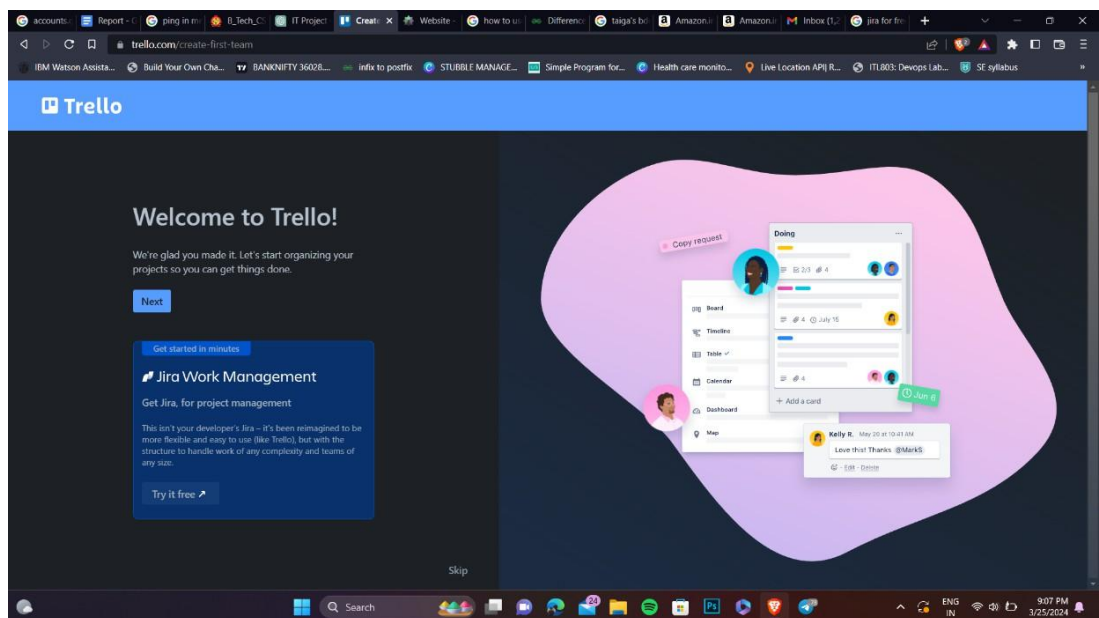
The individual activities to meet the project completion time.

AIM:

ALGORITHM:

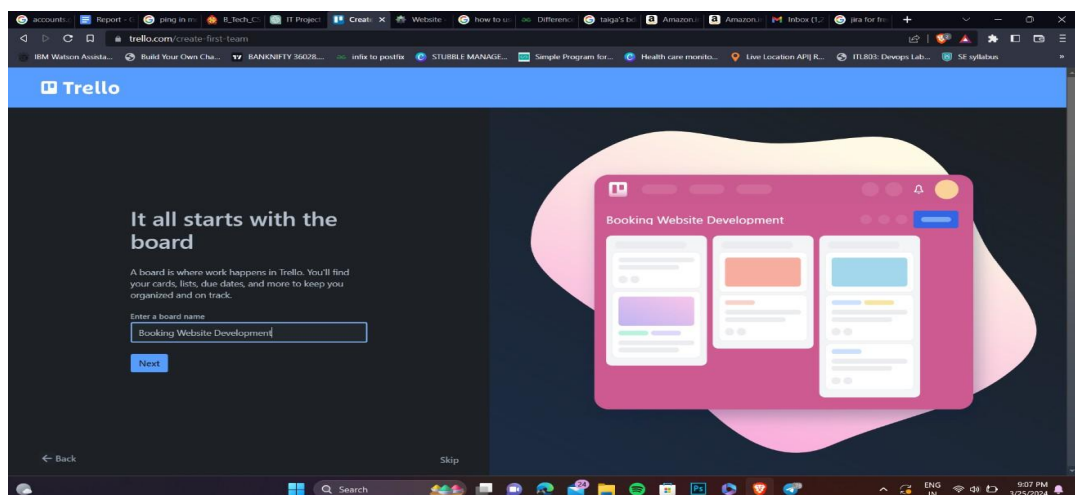
Step 1:

Create An Account in Trello an Project Organizer Tool



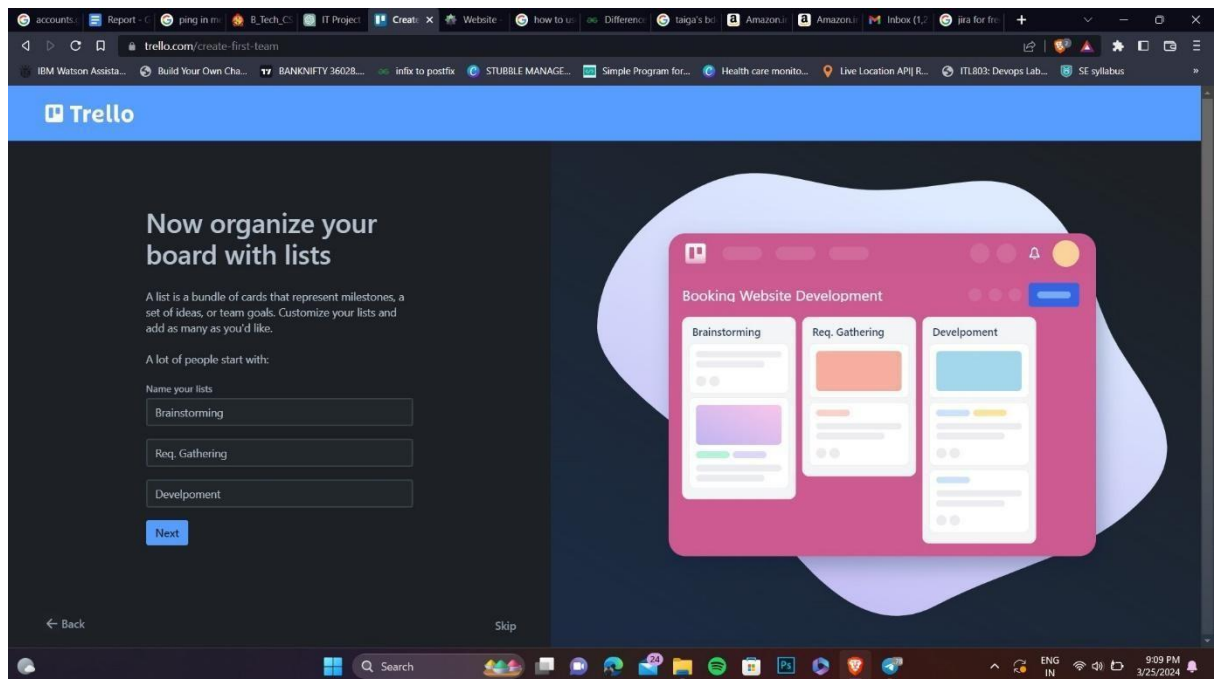
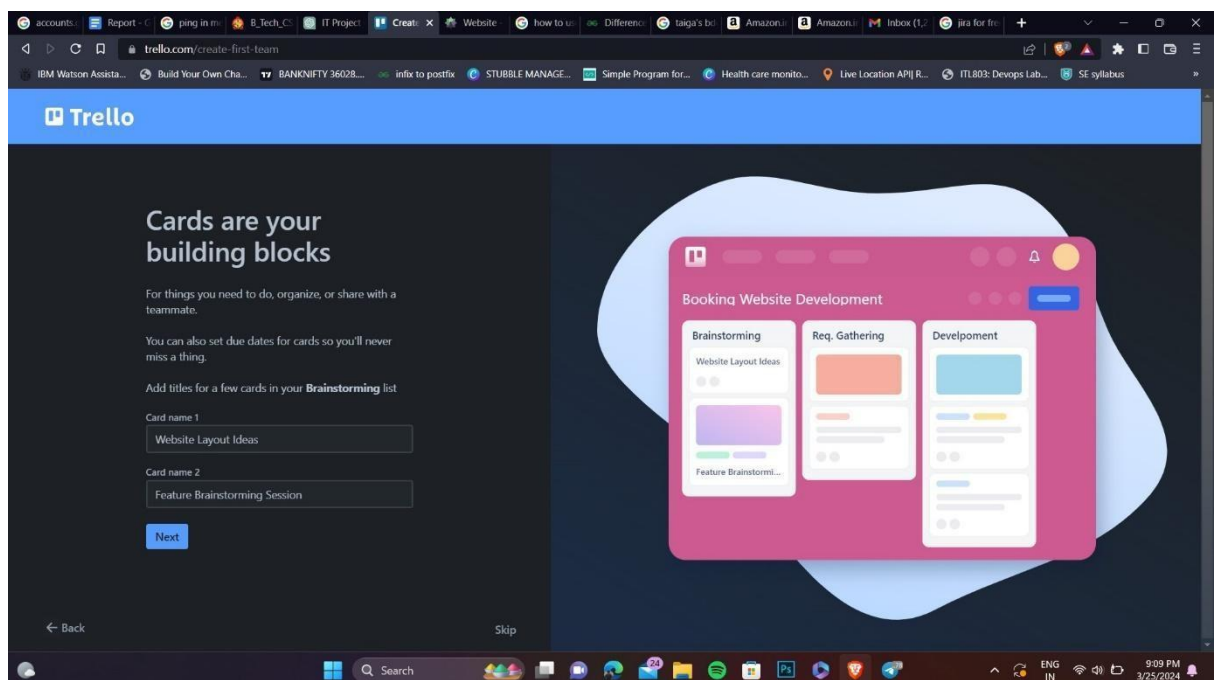
Step 2:

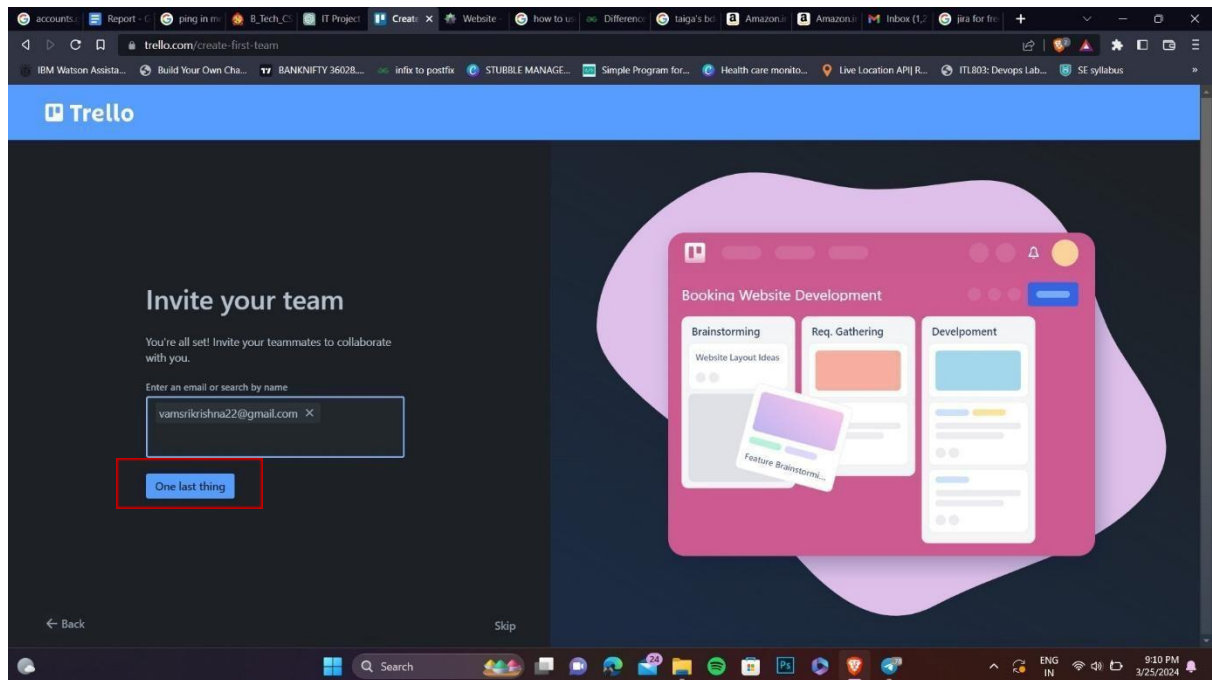
Name the project name as the Board Name



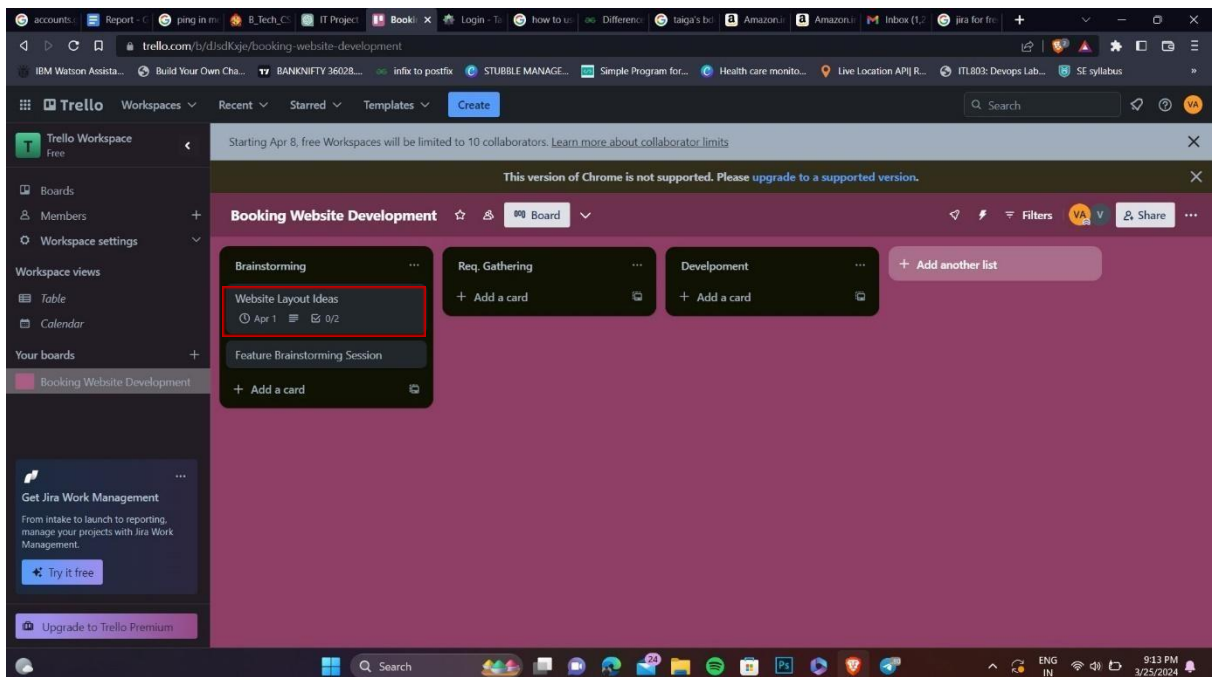
Step 3:

Name the list of Works to be done

**Step 4: Name the Card Name According to the project needs**

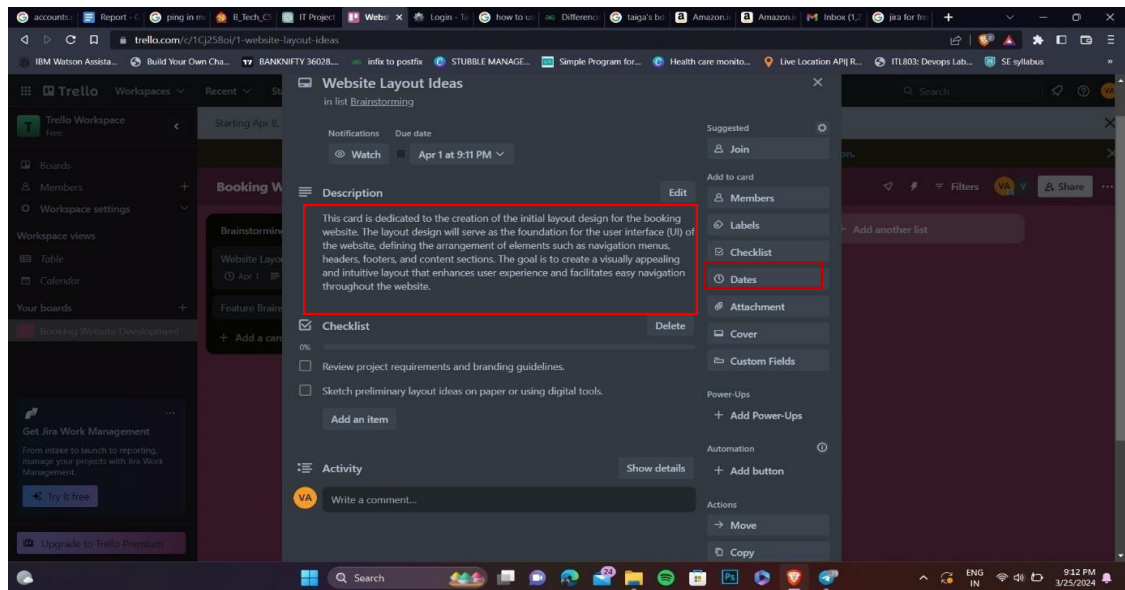
Step 5: Now invite the team members to whom you're going to Assign the Work

Click on Last Thing

Step 6: We Could View the Dashboard Of Our Project

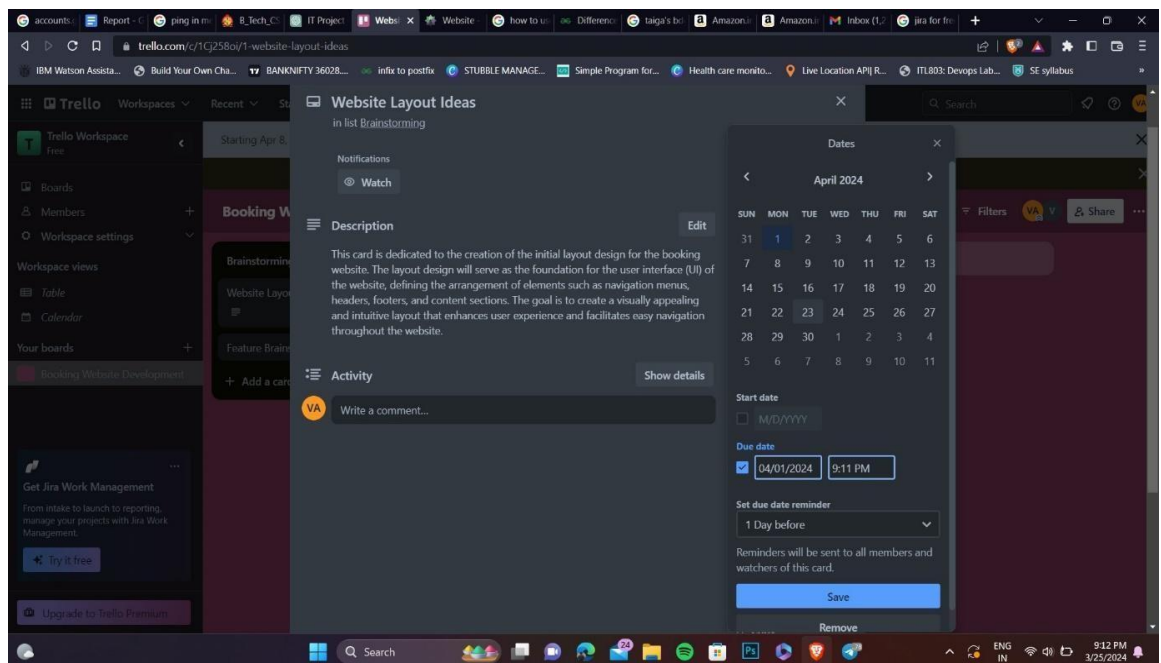
Now Click on the Website layout Ideas Card

Step 7: Now Enter the Description of Your Project Under the description area



And Click the Dates Button to add Due date for completion of time

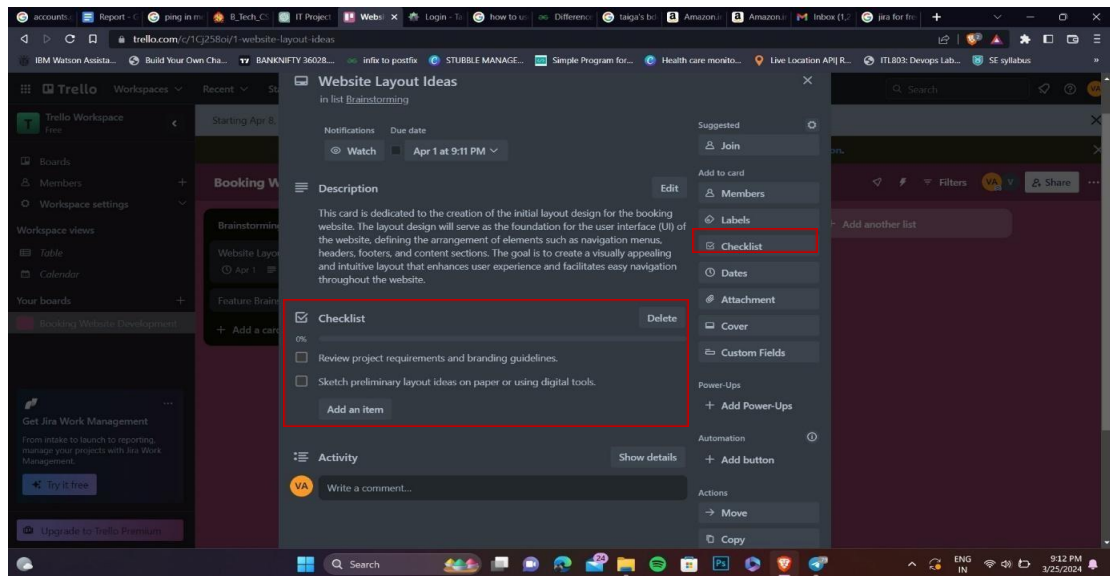
Step 8:



Now add the Due date for the developer

Step 9:

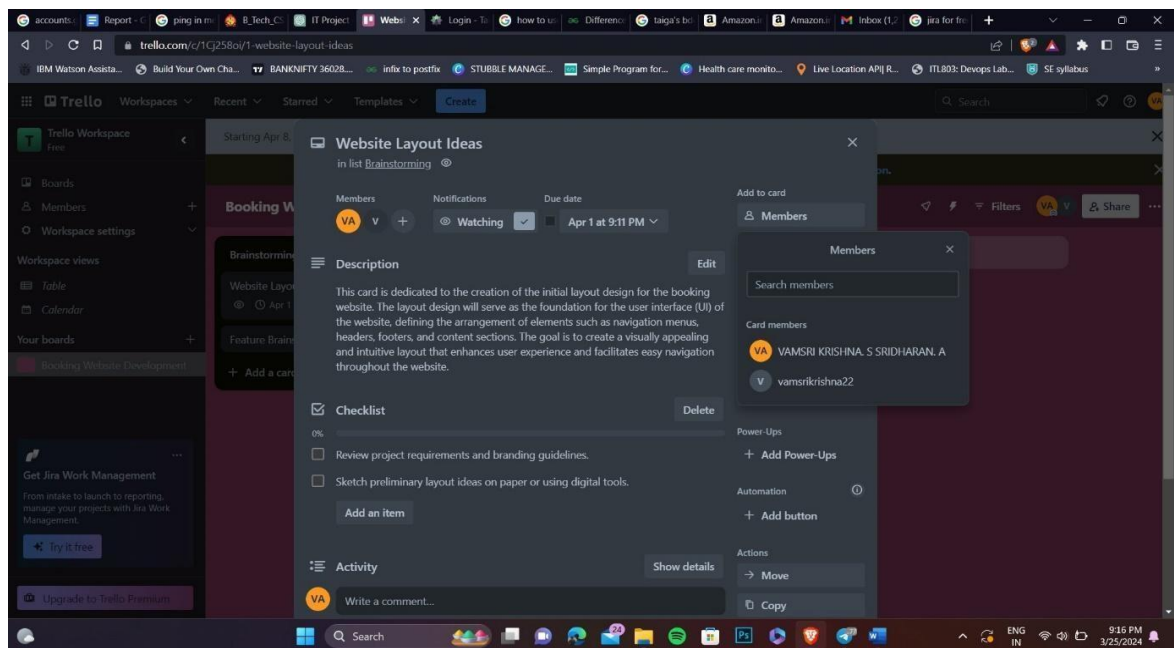
Now Click the Checklist to add the task to the user



We can view the checklist and click Members in the top

Step 10:

Now Assign the Member to the Project



Now the Task Will be Assigned to the Team member and It needs to be completed before due

RESULT:

Ex No: 5**Identify the critical bottleneck activities where any delays must be avoided to prevent delaying project completion .****AIM:****ALGORITHM:****MEASURES:****1. Review Project Plan:**

Start by reviewing the project plan, including the schedule, tasks, dependencies, and milestones.

2. Identify Key Dependencies:

Identify tasks or activities that have dependencies on other tasks or resources. These dependencies can create bottlenecks if not managed effectively.

3. Assess Resource Allocation:

Review resource allocation for each activity to ensure that sufficient resources are available to complete tasks on time. Resource constraints can lead to bottlenecks.

4. Analyze Task Durations:

Analyze the duration of each task or activity in the project plan. Tasks with long durations critical paths are potential bottleneck activities.

5. Consider External Dependencies:

Identify any external dependencies, such as third-party vendors or regulatory approvals, that may impact project timelines. Delays in obtaining external dependencies can cause bottlenecks.

6. Evaluate Critical Paths:

Identify critical paths in the project plan, which represent the sequence of tasks that determine the shortest duration for project completion. Any delays in tasks on the critical path will directly impact project completion.

7. Prioritize High-Risk Activities:

Prioritize activities with high risk or uncertainty, as these may have a higher likelihood of causing delays. Addressing risks proactively can help prevent bottlenecks.

8. Consider Mitigation Strategies:

Develop mitigation strategies for identified bottleneck activities. This may include allocating additional resources, adjusting task priorities, or implementing contingency plans.

9. Monitor and Track Progress:

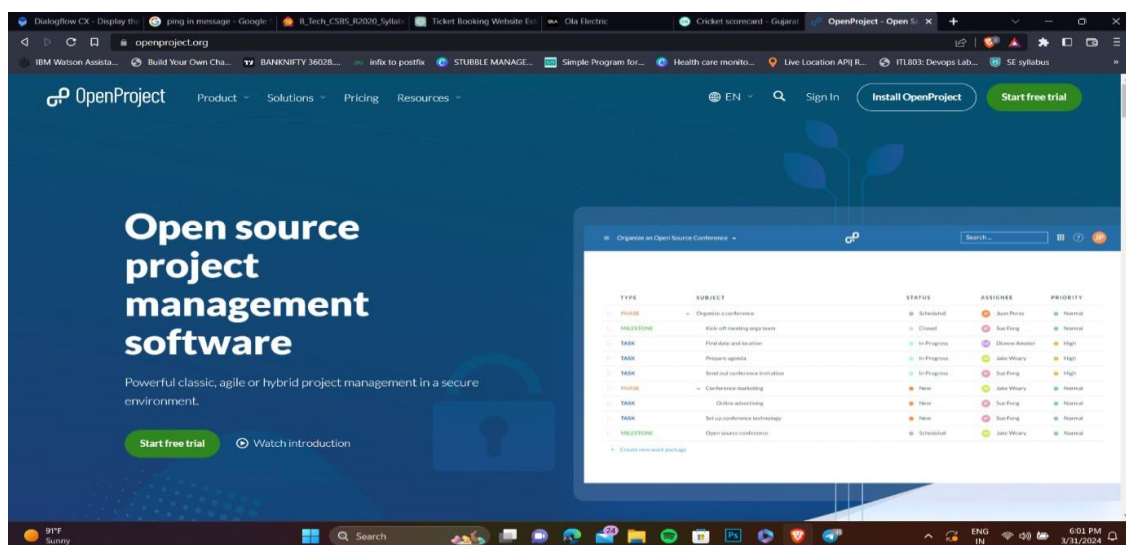
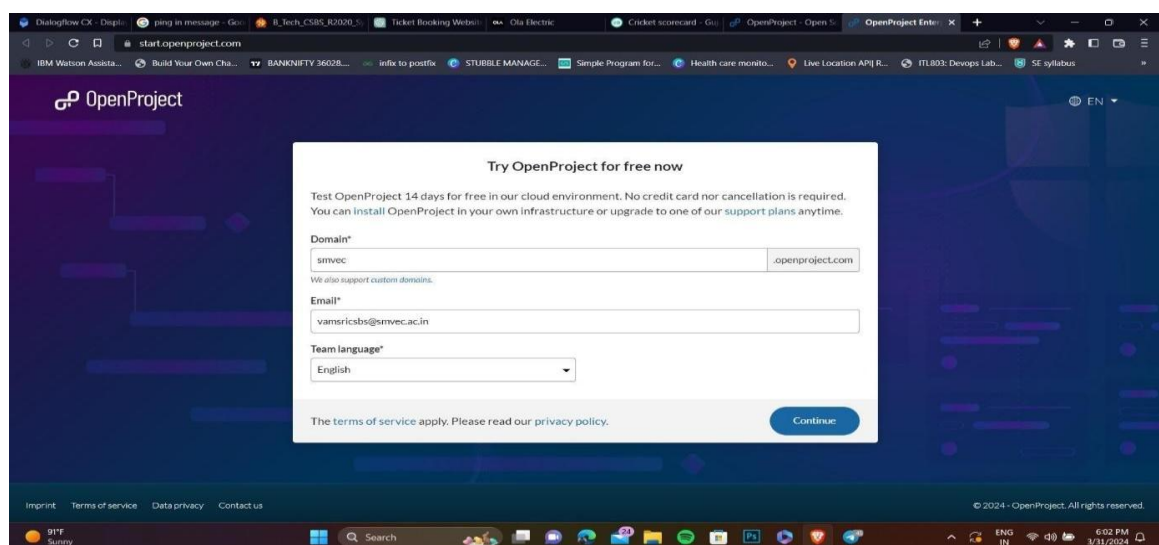
Continuously monitor and track the progress of critical bottleneck activities throughout the project lifecycle. Early identification of potential delays allows for timely intervention and mitigation.

10. Regular Communication:

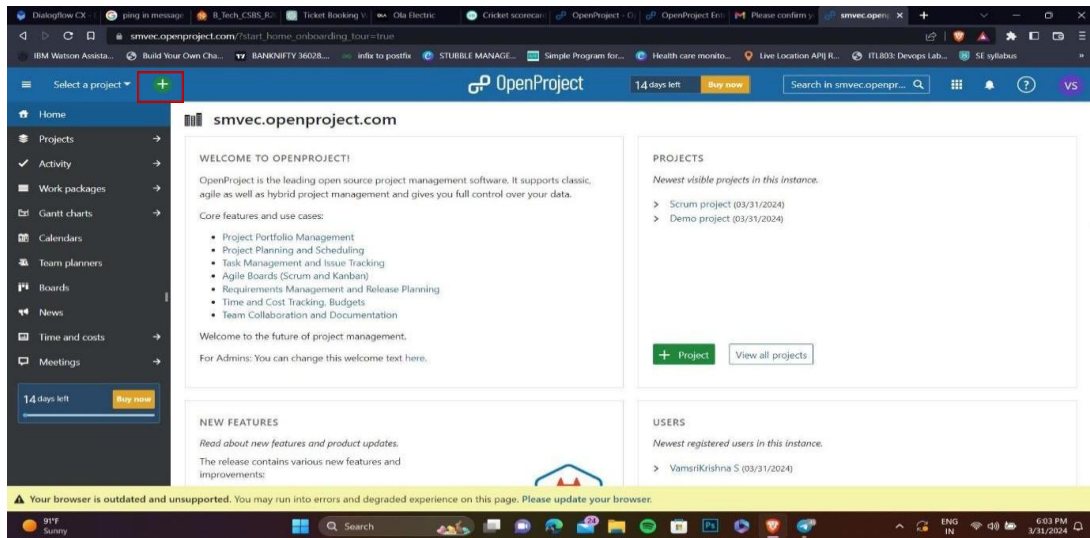
Maintain open communication with project team members and stakeholders to ensure awareness of critical bottleneck activities and any potential risks or issues.

RESULT:

Ex.No: 6

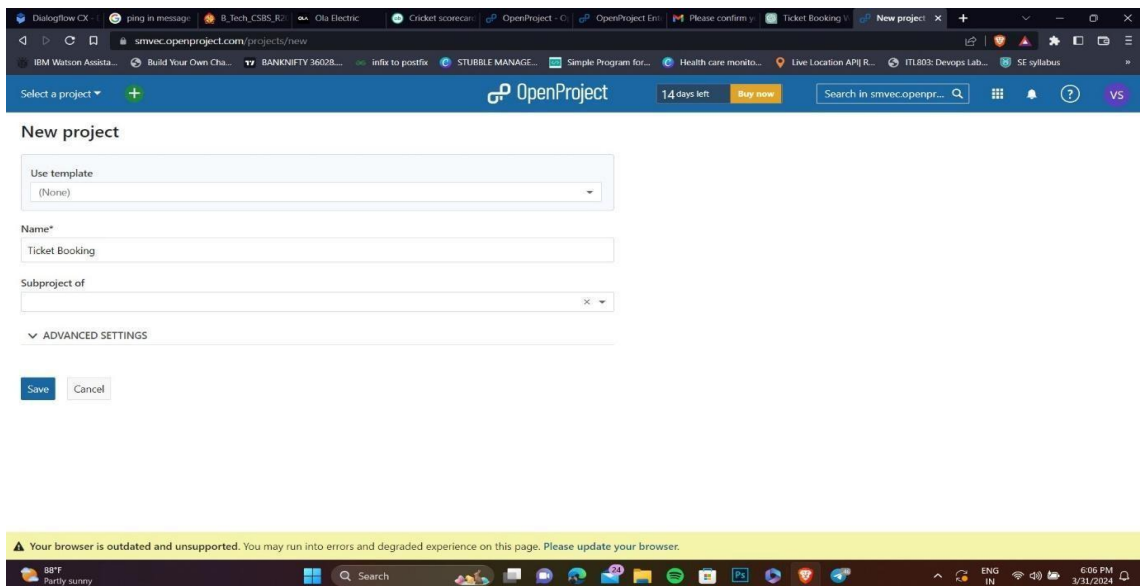
**IT project risk analysis (includes planning during uncertainty)
using open-source tools****AIM:****ALGORITHM:****Step 1:**Open your web browser and navigate to the OpenProject website**Step 2:** Enter the Domain name as SMVEC, Enter Your email ID

Once After Confirming with your email ID, You will be viewing the home screen

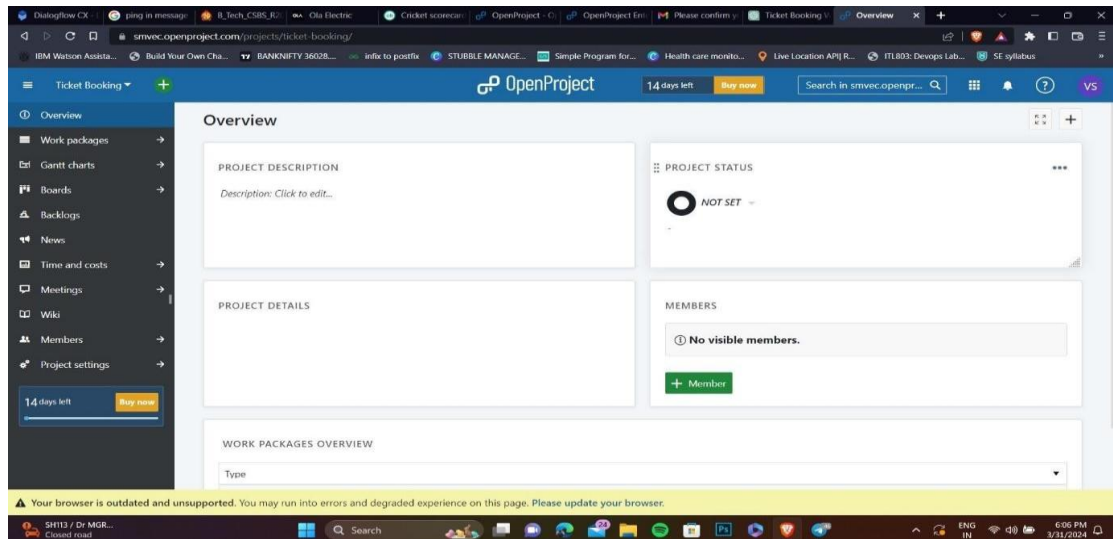


Click the New button at the top and create New project

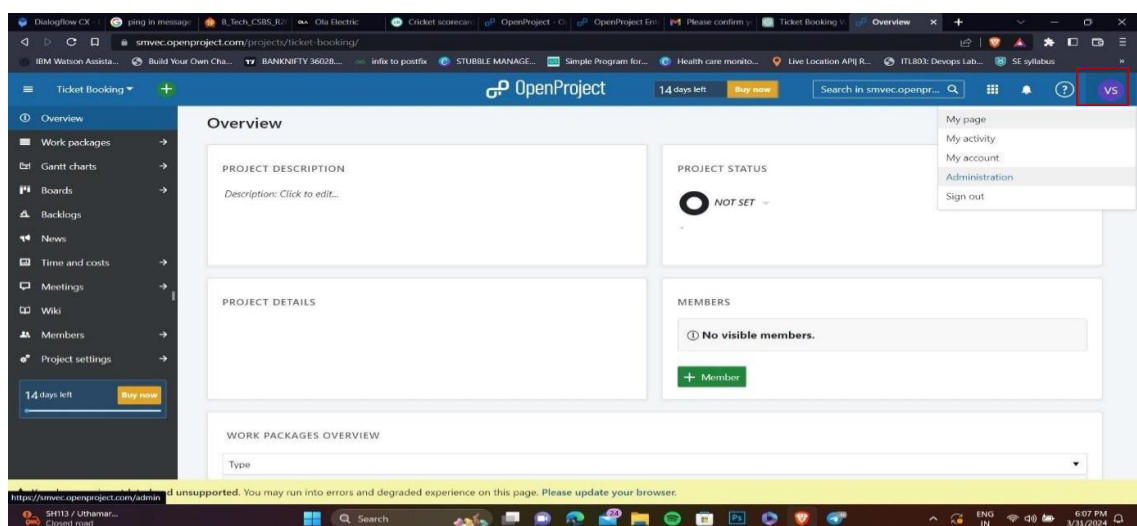
Step 3: Enter the Project Name as Ticket Booking



Step 4: We can view the Project Overview Tab

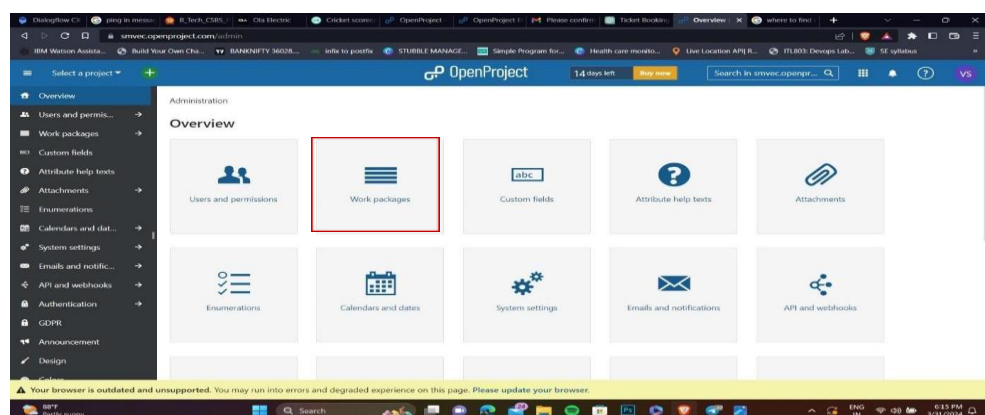


Step 5: Click On the Profile Button on the top right



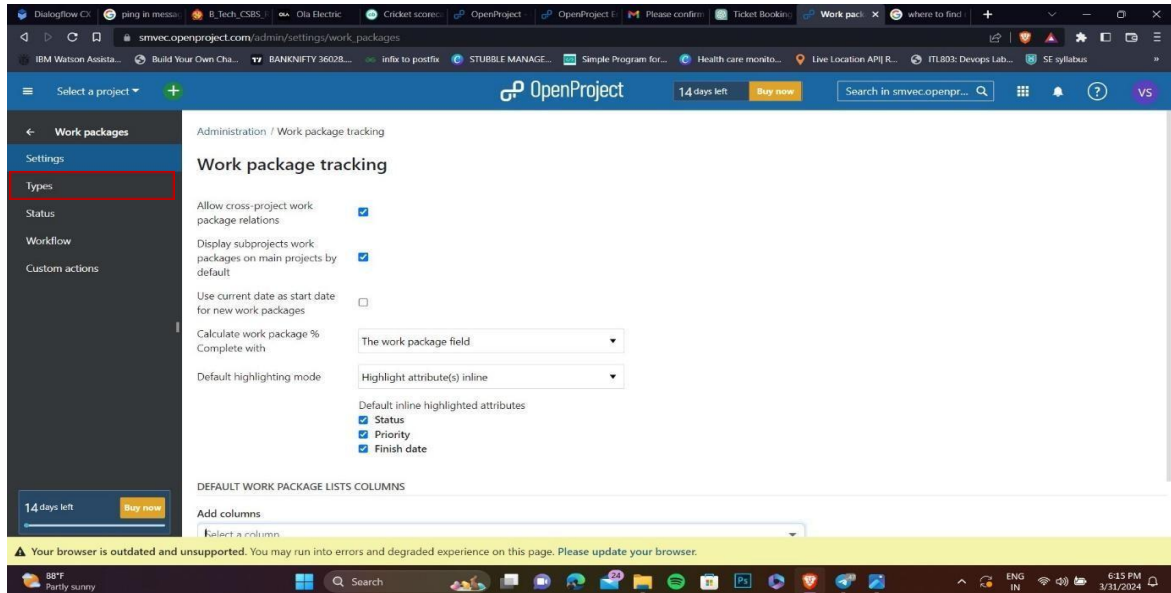
And Click the Administration

Step 6: We can View the Administration Tab

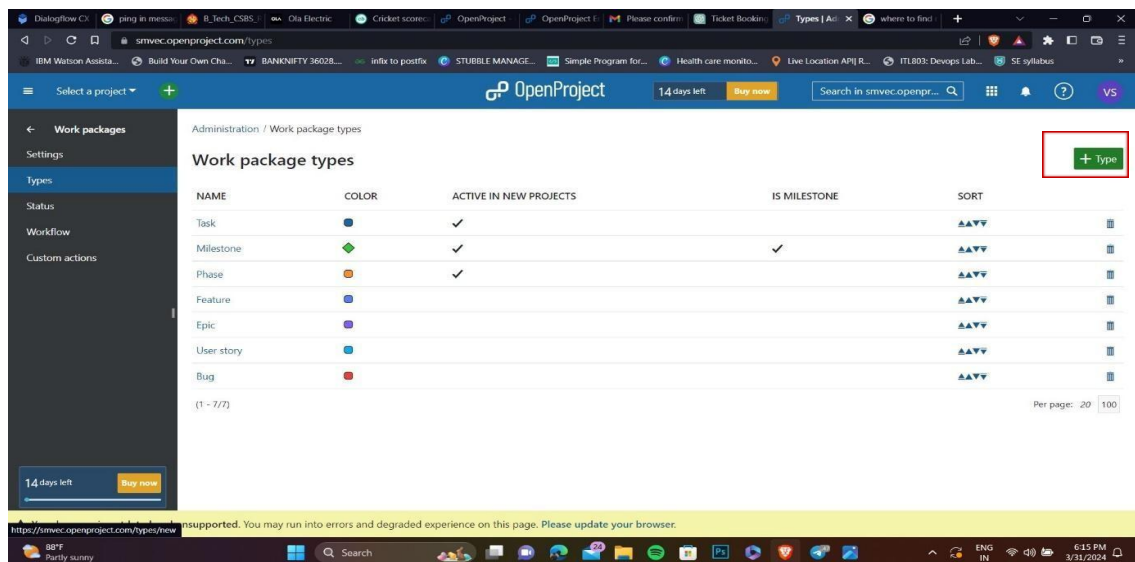


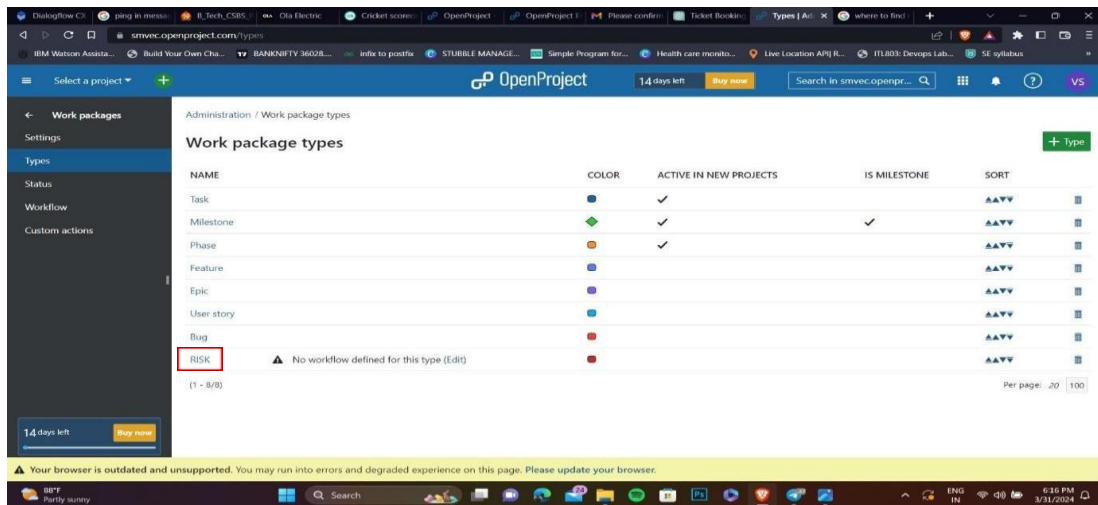
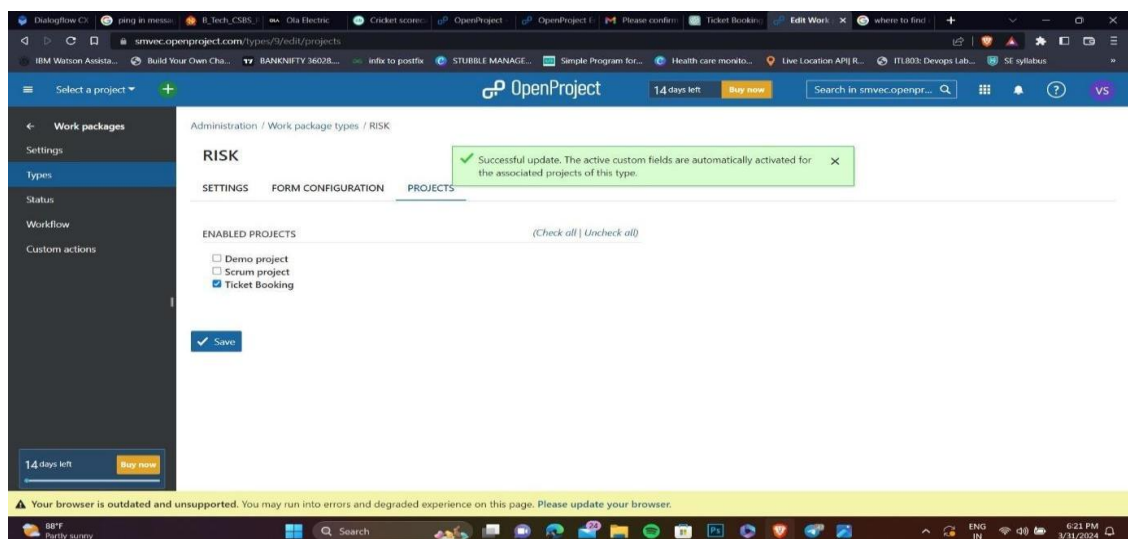
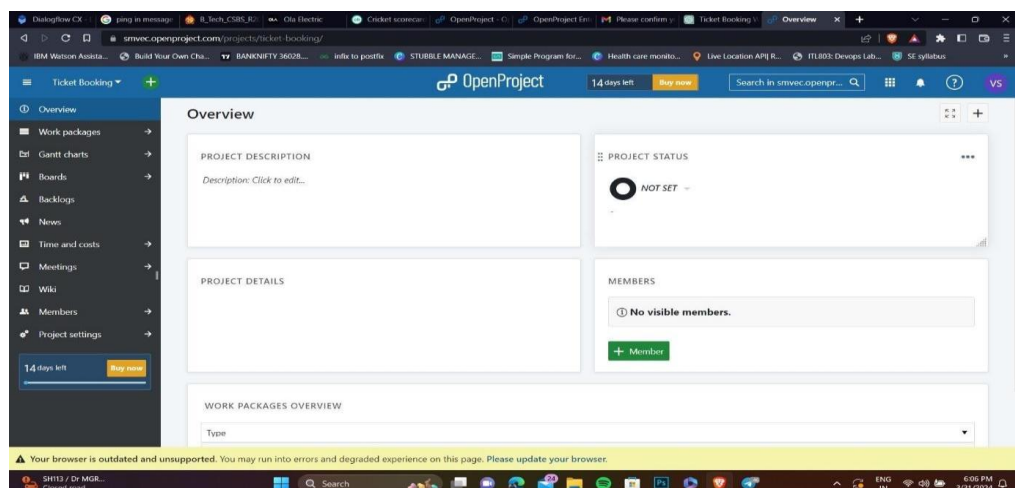
Now Click on Work Packages

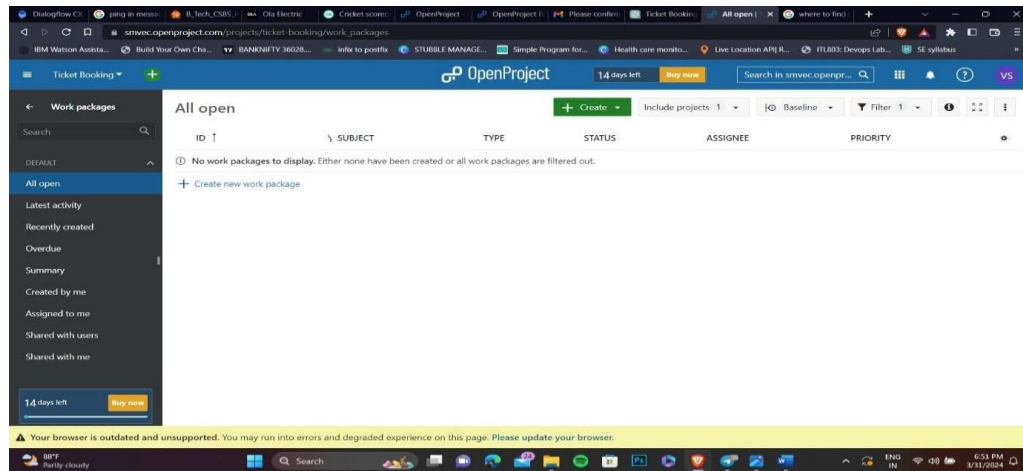
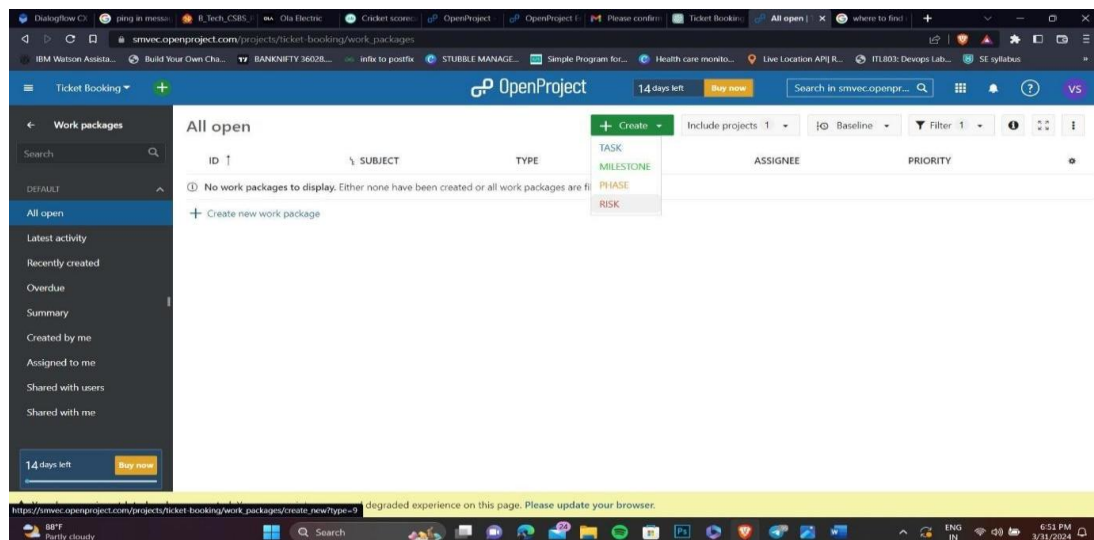
Step 7: Now Click Types in the left menu



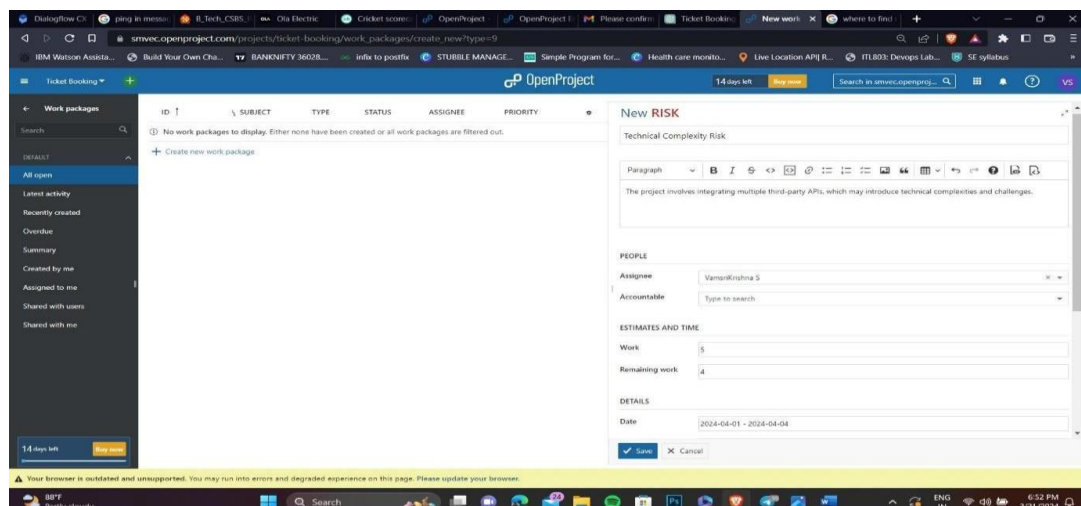
Step 8: Click Type in the top right and Name it as RISK and Assign the Colour as RED and S Save.



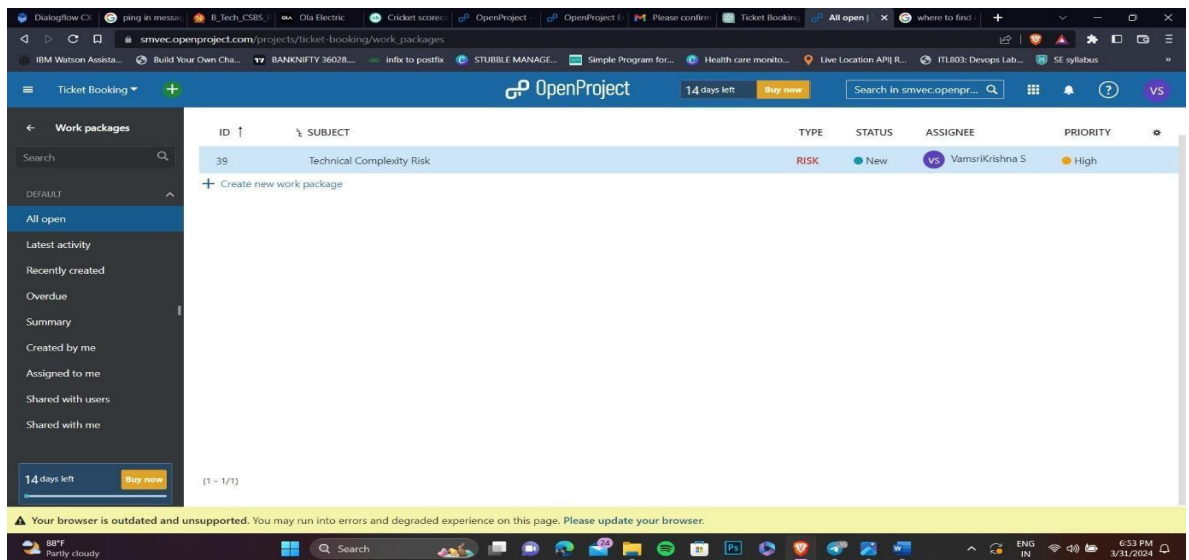
Step 9: Now Click the Risk in types tab**Step 10: Under Projects tab enable Ticket Booking and Save it. Now Navigate to Ticket Booking Project by Select a project Ticket Booking****Step 11: Click Work Packages**

Step 12: Click create at the top**Step 13: Choose the type as RISK****Step 14: Enter the RISK Name, Description, Assignee, due According to the need of your project and Make Priority as HIGH**

And click Save



Step 15: Now We can view the Riskiest Task based on the analysis and it will be given the top most priority



RESULT:

Ex.No: 7	Design IT Project Audit Template	
AIM: PROCEDURE: Project Overview: - Project Name: Appointment Booking Experiment - Project Manager: JOHN - Start Date: March 20, 2024 - End Date: April 14, 2024 - Audit Date: April 20, 2024 - Auditor(s): [Jolly Joseph] 1. Project Planning: - [X] Was the experiment scope clearly defined? (Yes) - [X] Were experiment objectives and success criteria well-defined? (Yes) - [X] Were resources (human, financial, technological) adequately allocated for the experiment? (Yes) - [X] Was there a detailed experimental plan with milestones and timelines? (Yes) 2. Experiment Execution: - [X] Were tasks and responsibilities clearly assigned? (Yes, roles were clearly defined in the project plan) - [X] Did the experiment team adhere to the planned schedule? (Yes, the project timeline was followed closely) - [X] Were there any deviations from the original experiment plan? If so, were they justified? (To be filled - If any deviations occurred, they will be documented here along with justifications) - [X] Were experiment risks identified and managed effectively? (To be filled - Risks were identified but require further analysis) - [X] Were experiment changes documented and communicated appropriately? (To be filled - Any changes will be documented here)		

3. Quality Assurance:

- [X] Was there a defined quality assurance process for the experiment? (To be filled - QA process was outlined but needs further evaluation)
- [X] Were quality standards met throughout the experiment? (To be filled - Standards were established but require verification)
- [X] Were there mechanisms in place for testing and debugging the booking system? (To be filled - Testing mechanisms were implemented but need assessment)
- [X] Were user acceptance tests conducted satisfactorily? (To be filled - Testing is underway and results are pending)
- [X] Were issues and defects tracked and resolved promptly? (To be filled - Issue tracking system was in place, results pending)

4. Stakeholder Communication:

- [X] Was there effective communication with stakeholders throughout the experiment? (To be filled - Communication channels were established but require evaluation)
- [X] Were experiment progress updates provided regularly? (To be filled - Updates were scheduled but require assessment)
- [X] Were stakeholders involved in decision-making processes related to the experiment? (To be filled - Stakeholder involvement was encouraged but requires verification)
- [X] Were concerns and issues addressed in a timely manner? (To be filled - Issues were logged but require evaluation of resolution timeliness)

RESULT:

Ex.No: 8

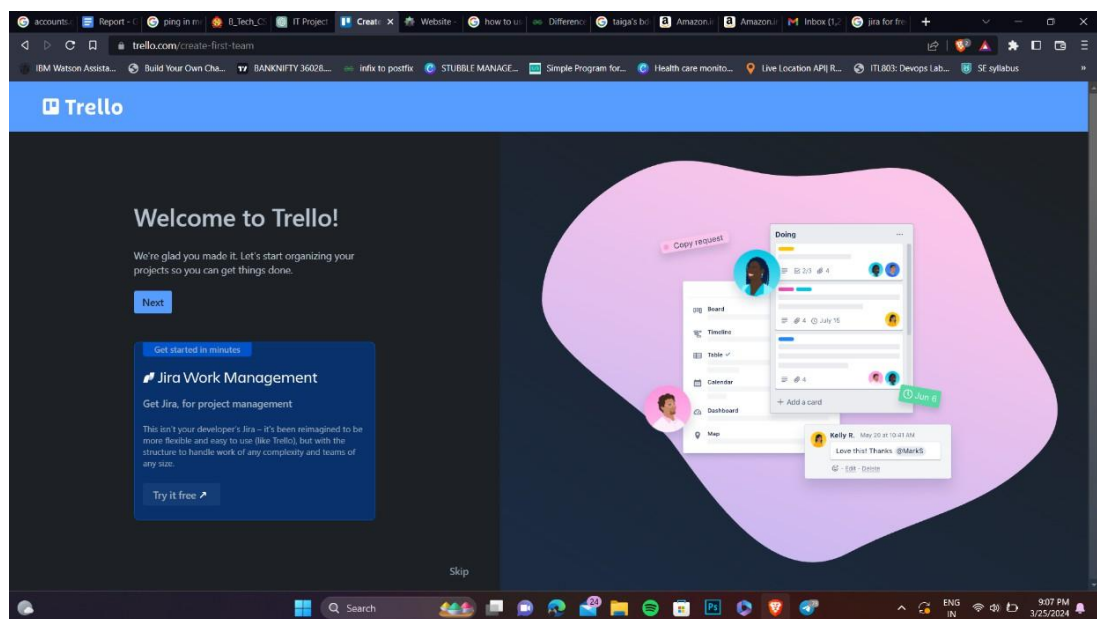
Agile Project Management Tools (Open source)

AIM:

ALGORITHM:

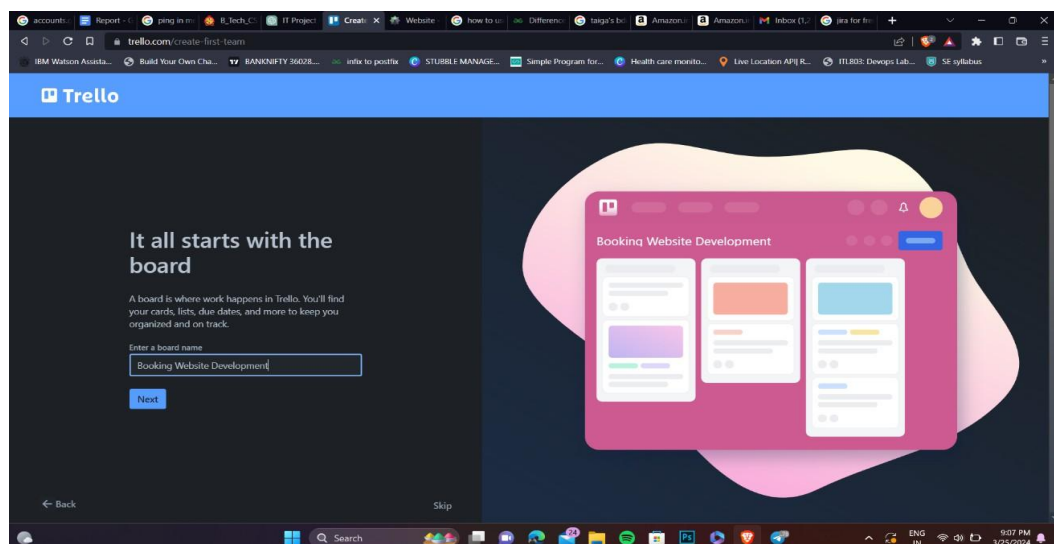
Step 1:

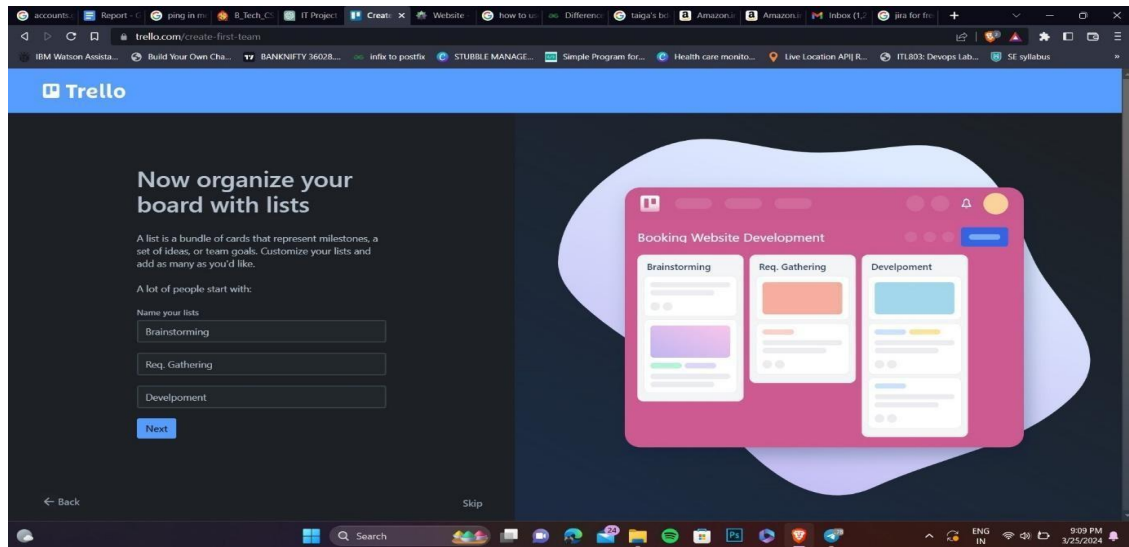
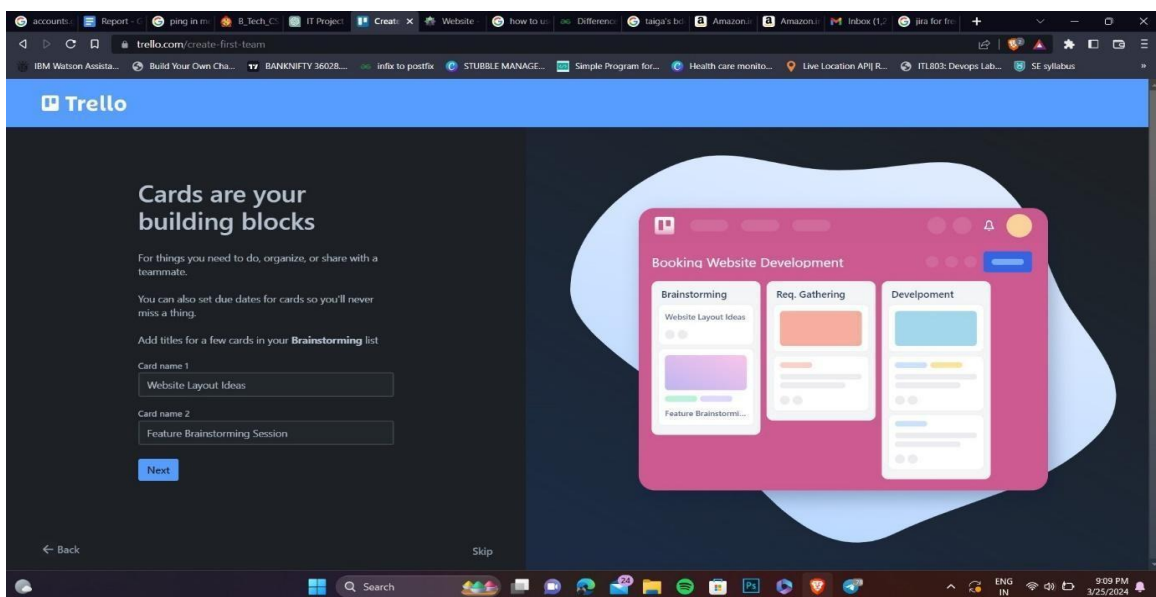
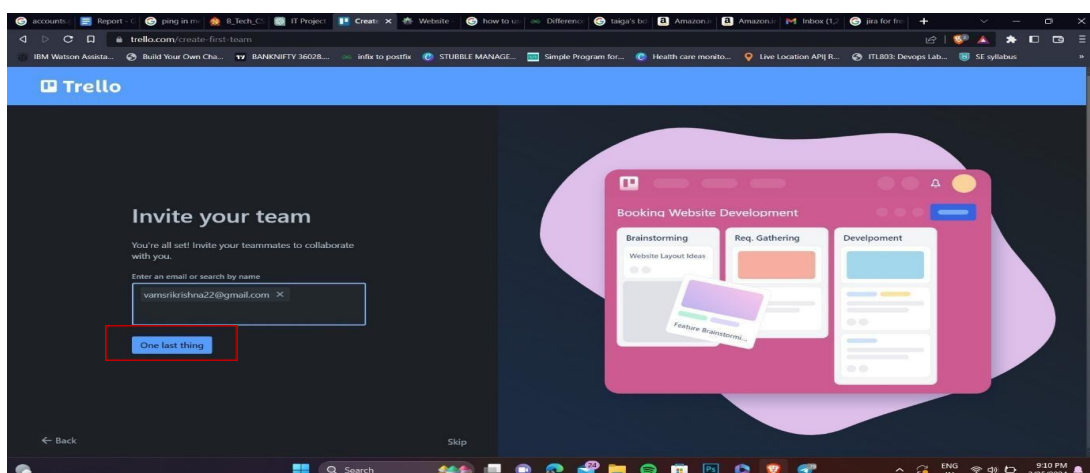
Create An Account in Trello an Project Organizer Tool



Step 2:

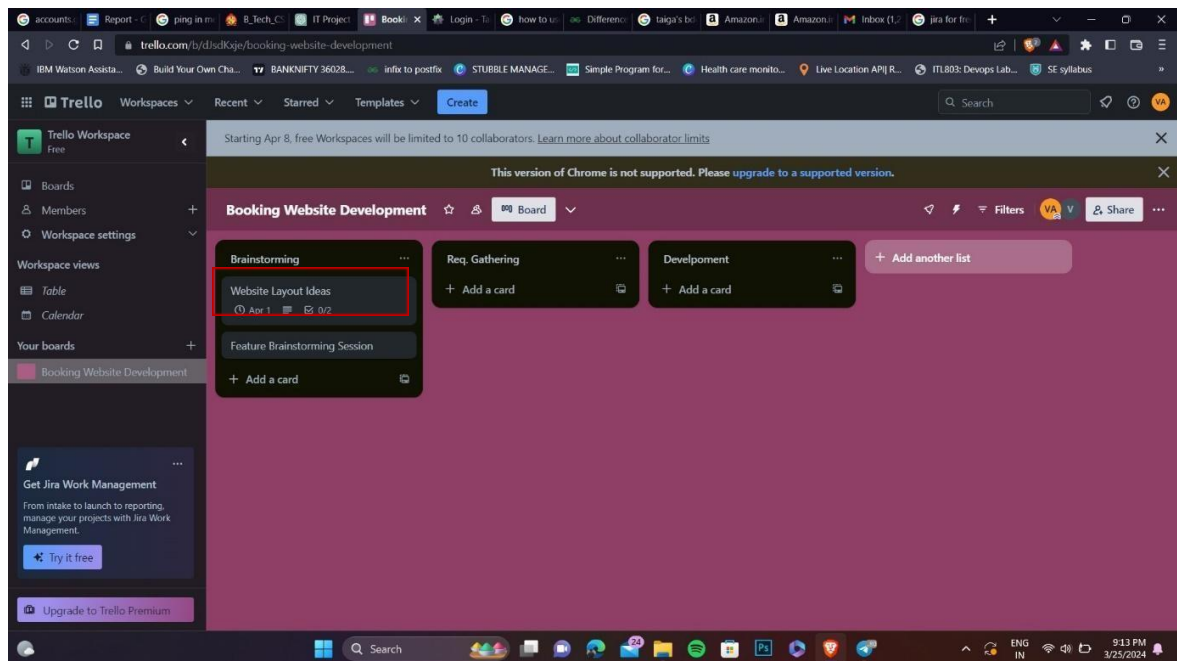
Name the project name as the Board Name



Step 3: Name the list of Works to be done**Step 4: Name the Card Name According to the project needs****Step 5: Now invite the team members to whom you're going to Assign the Work**

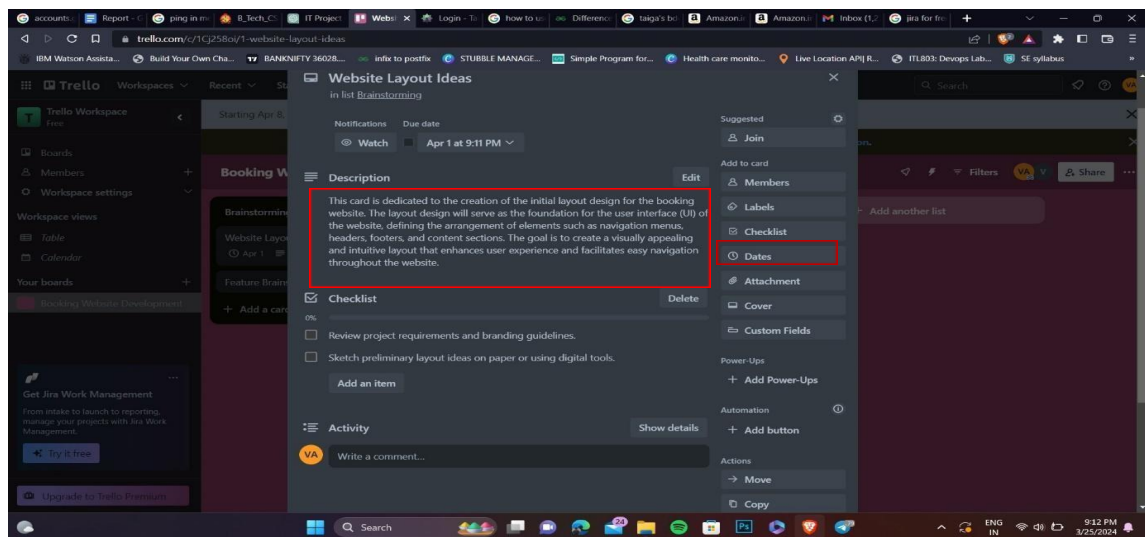
Click on Last Thing

Step 6: We Could View the Dashboard Of Our Project



Now Click on the Website layout Ideas Card.

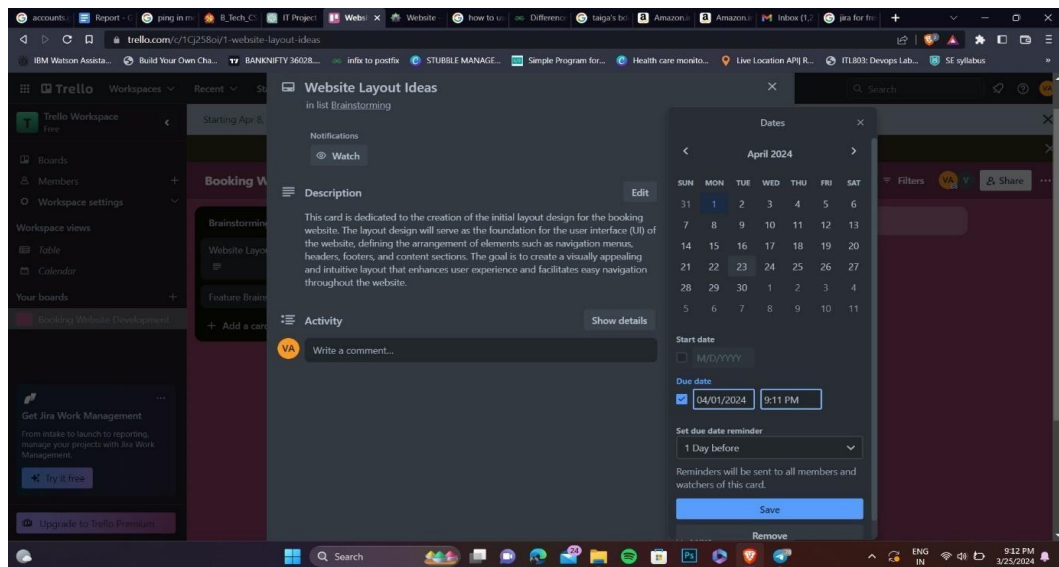
Step 7: Now Enter the Description of Your Project Under the description area



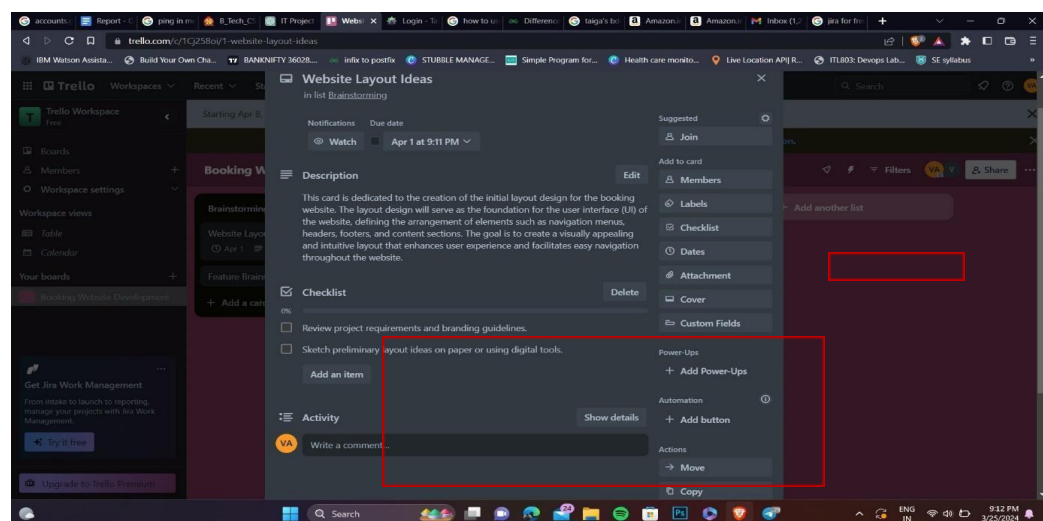
And Click the Dates Button to add Due date for completion of time.

Step 8:

Now add the Due date for the developer

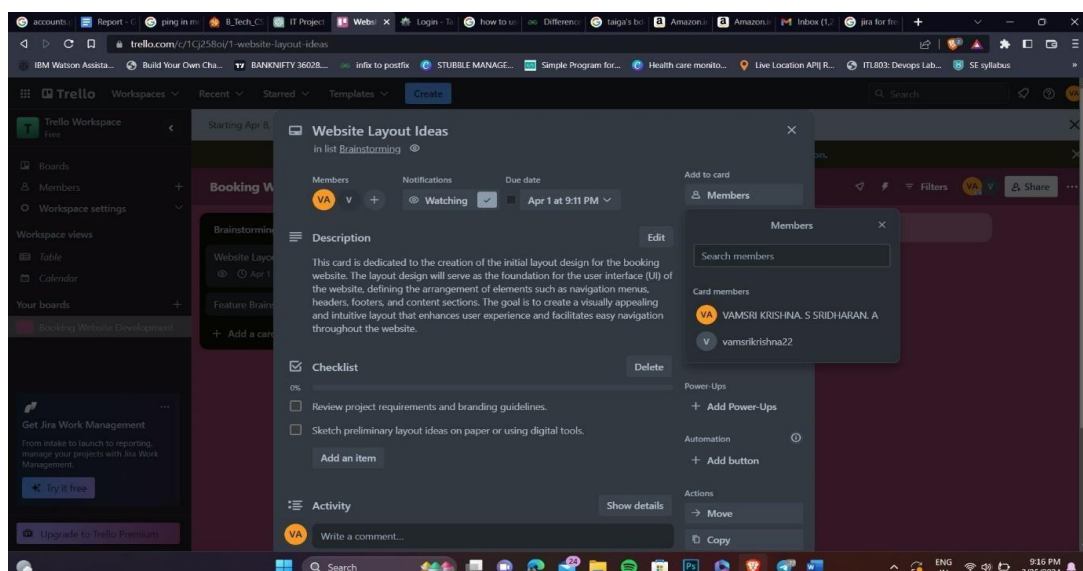


Step 9: Now Click the Checklist to add the task to the user



We can view the checklist and click Members in the top Step

Step10: Now Assign the Member to the Project



Now the Task Will be Assigned to the Team member and It needs to be completed before due
This is one of the AGILE Approaches for BOOKING WEBSITE where in the card we could see the iterative approach.

RESULT:

Ex.No: 9

Design IT Service Management (ITIL) Templates

AIM:**PROCEDURE:****INCIDENT MANAGEMENT TEMPLATE:****Incident ID:** IM-2024-001**Date Reported:** March 30, 2024**Description of Incident:**

A system error was encountered when users attempted to submit claims through the online portal. Users reported receiving an error message stating "Internal Server Error" upon submission.

Impact Assessment:

The incident has a high impact on business operations as it prevents users from submitting claims online, leading to potential delays in claims processing and customer dissatisfaction.

Priority Classification:

High

Assigned Technician: Prithvi

Support Group: IT Operations

Status Updates:

Date/Time: March 30, 2024 10:00 AM

Update: Initial investigation initiated by the IT Operations team to identify the cause of the error. Date/Time: March 30, 2024 11:30 AM

Update: Identified a server configuration issue causing the Internal Server Error. Planning resolution steps.

Date/Time: March 30, 2024 1:00 PM

Update: Applied configuration fix on the server. Testing the claim submission functionality to ensure resolution.

Resolution Details:

The server configuration issue causing the Internal Server Error has been successfully resolved. Users can now submit claims through the online portal without encountering errors.

Root Cause Analysis:

Root cause analysis identified a misconfiguration in the server settings, leading to the Internal Server Error. The misconfiguration occurred during a recent system update.

RESULT:

Ex No: 10

Scrum: IT Project Management, DevOps and Automated Testing Tools

Scrum, a popular Agile framework, emphasizes iterative development, continuous feedback, and self-organizing teams. When combined with IT project management, DevOps practices, and automated testing tools, it enables efficient delivery of high-quality software products. Here are some tools that can be used in each area:

1. IT Project Management:

-Jira: Jira is a widely used project management tool that offers features for sprint planning, backlog management, task tracking, and issue prioritization. It allows teams to organize and manage their work using Scrum or Kanban boards.

- Trello: Trello is a visual collaboration tool that uses boards, lists, and cards to organize and prioritize tasks. It's suitable for smaller teams or projects and provides flexibility in managing tasks and workflows.

- Asana: Asana is a project management tool that offers features for task tracking, team collaboration, and project planning. It allows teams to create projects, assign tasks, set deadlines, and track progress in a centralized platform.

2. DevOps:

- Jenkins: Jenkins is a popular open-source automation server used for continuous integration and continuous delivery (CI/CD). It automates the build, test, and deployment processes, allowing teams to deliver software updates more frequently and reliably.

- GitLab CI/CD: GitLab CI/CD is an integrated part of the GitLab platform, providing built-in CI/CD capabilities for automating the software delivery pipeline. It allows teams to define, test, and deploy code changes using a single, integrated platform.

- CircleCI: CircleCI is a cloud-based CI/CD platform that automates the testing and deployment process. It supports integration with popular version control systems like GitHub and Bitbucket and allows teams to build, test, and deploy code changes automatically.

3. Automated Testing Tools:

- Selenium: Selenium is an open-source testing framework used for automating web application testing across different browsers and platforms. It allows teams to write automated tests using various programming languages like Java, Python, or C#, and execute them on different browsers.

- JUnit/TestNG: JUnit and TestNG are popular testing frameworks for Java applications. They provide annotations and assertions for writing unit tests, integration tests, and end-to-end tests, helping teams ensure the quality and reliability of their code.

- Cypress: Cypress is a modern JavaScript testing framework focused on end-to-end testing for web applications. It provides a comprehensive set of tools for writing and running automated tests, including built-in assertions, mocking, and spying functionalities.

By integrating these tools into the Scrum workflow, teams can streamline their development processes, improve collaboration, increase productivity, and deliver high-quality software products efficiently.

